



# **GEO 330 / SUD 430**

## **SUSTAINABLE URBAN TRANSPORTATION**

# *Week #8 agenda*

- *Administrative updates*
- *Evolution and impact of automobility and autonomous technologies*
- *Group project resources*
- *Group project breakout session*
- *Exercises #3, #4*
- *Course Q & A*



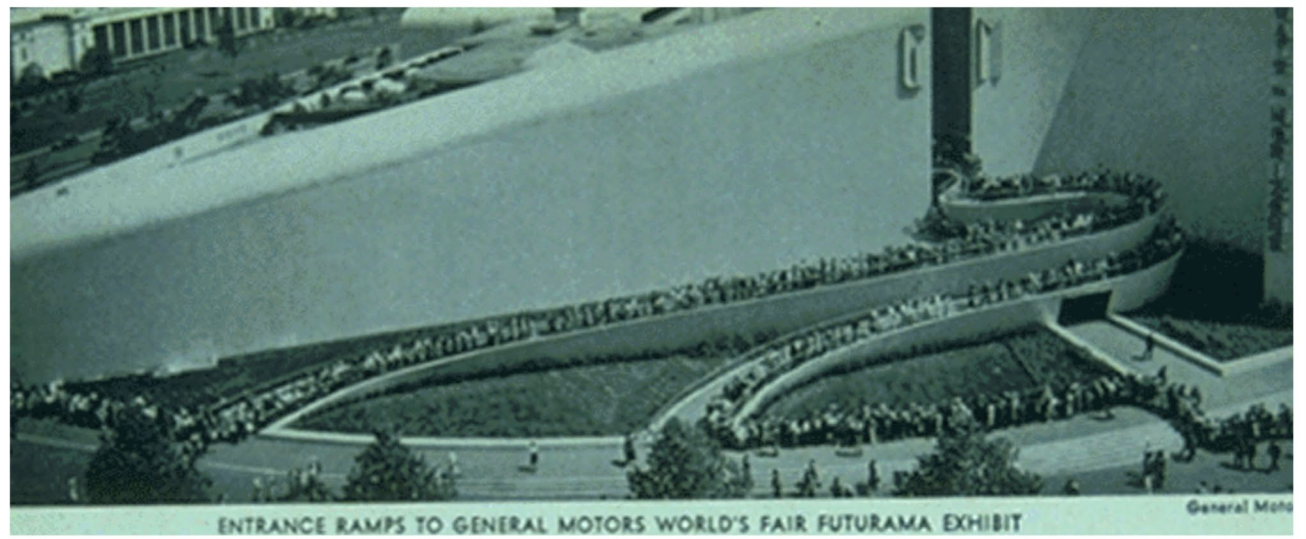
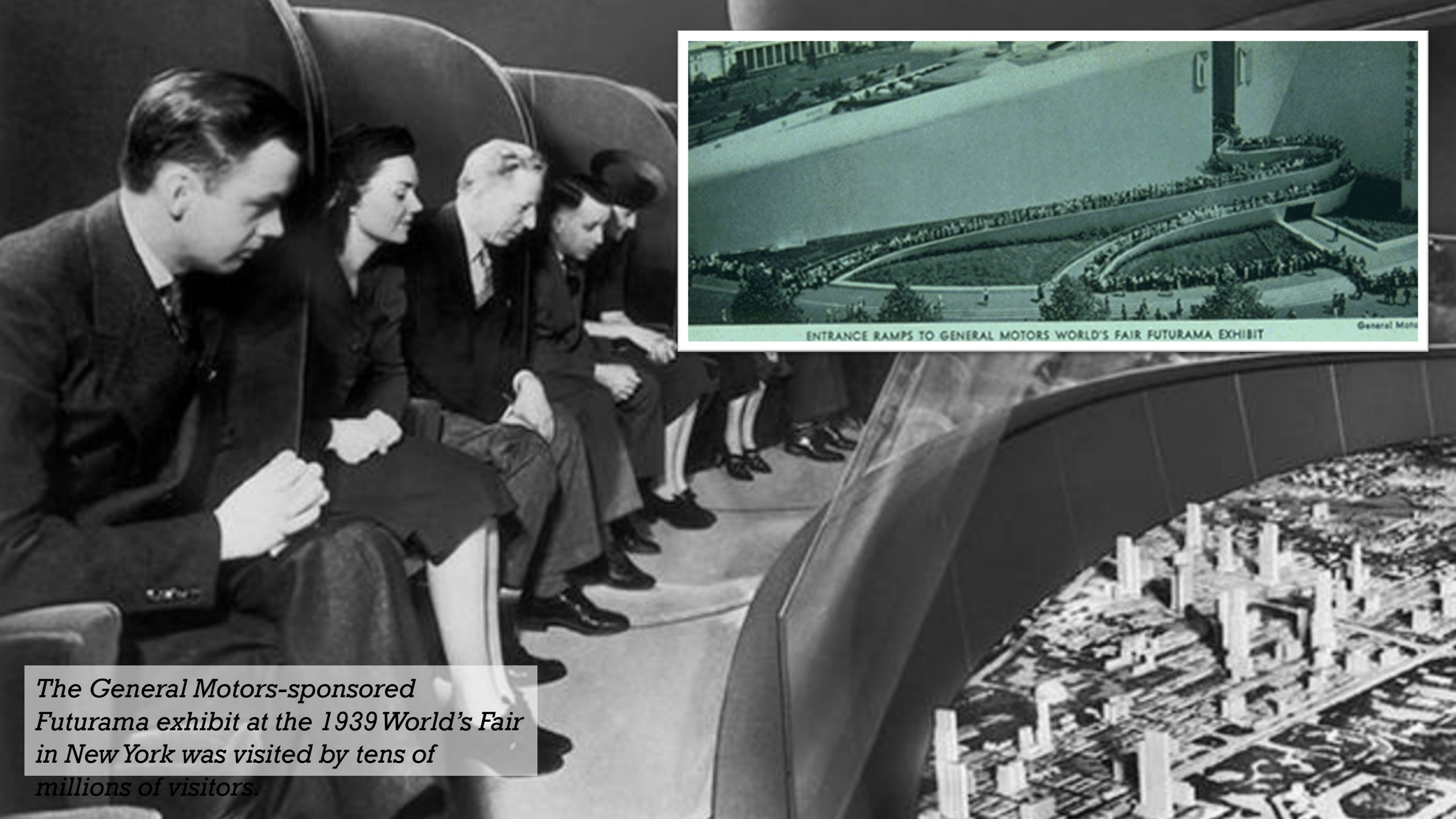


Lake Shore Drive, 1905  
[top]  
State Street 1910 [below]



South Water Street, 1890 [top]  
Chicago's "L" Elevated Train along Wabash St, 1900  
[below]





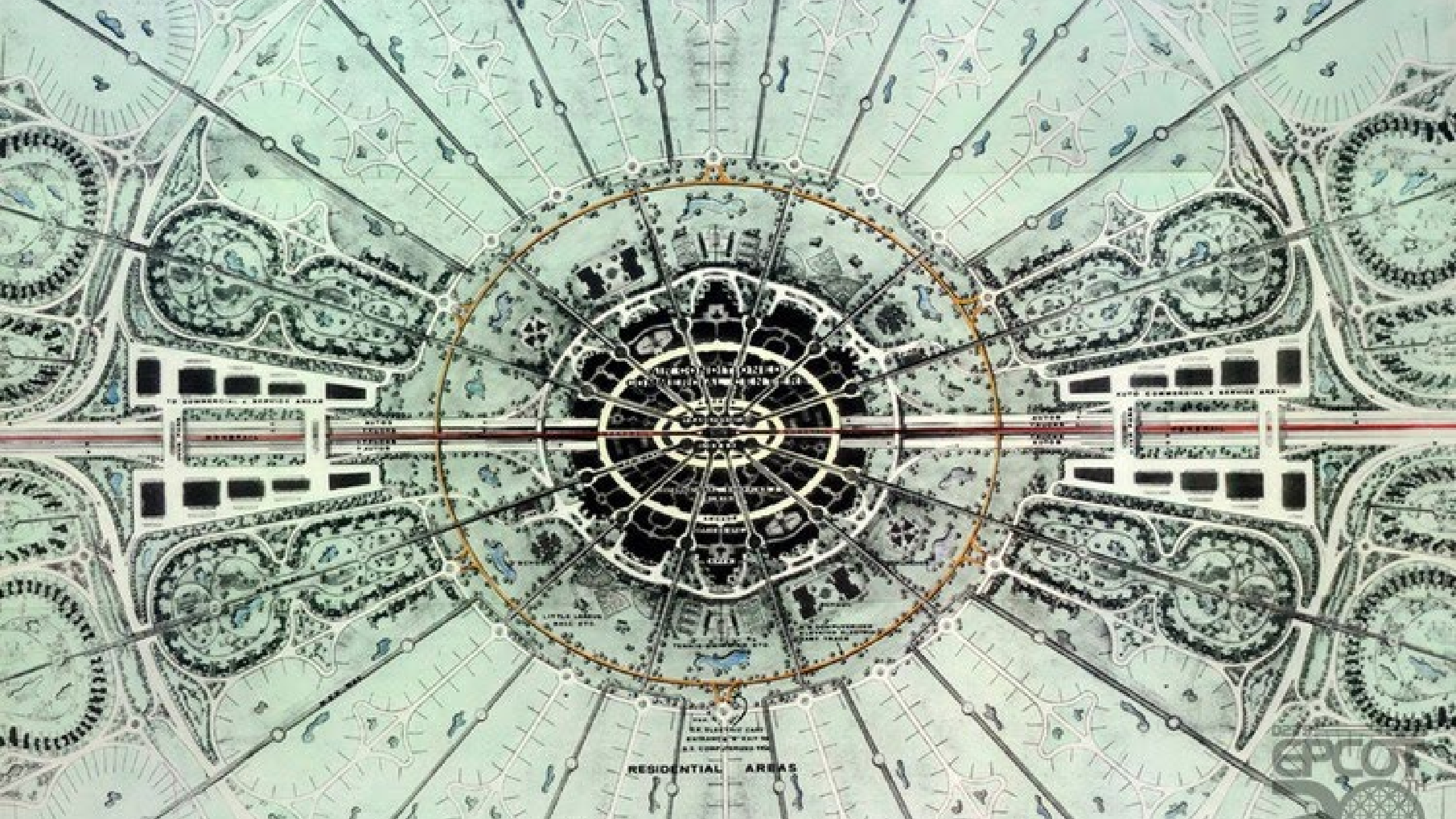
ENTRANCE RAMPS TO GENERAL MOTORS WORLD'S FAIR FUTURAMA EXHIBIT

General Motors

*The General Motors-sponsored  
Futurama exhibit at the 1939 World's Fair  
in New York was visited by tens of  
millions of visitors*

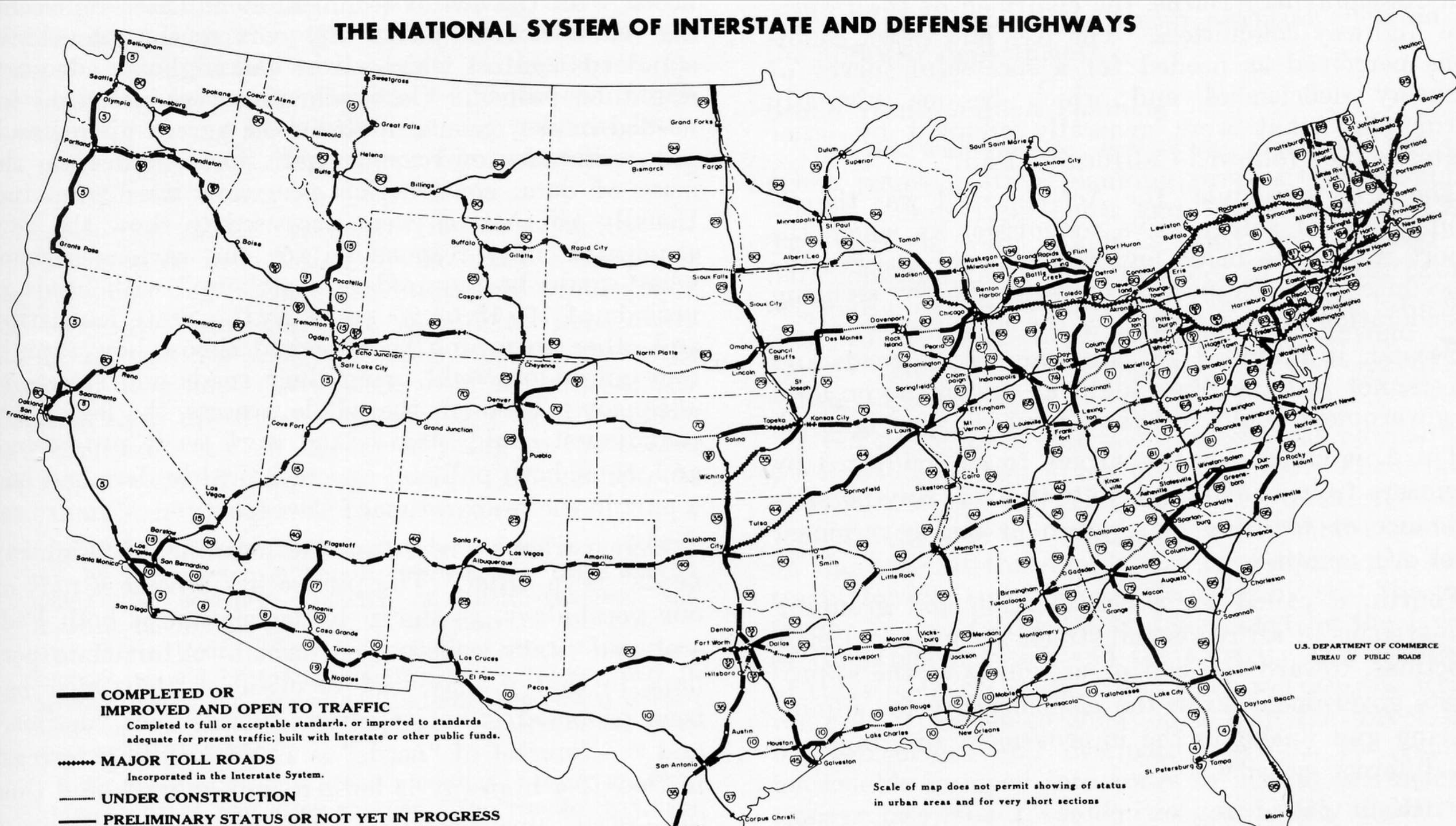
*“Residential, commercial, and industrial areas  
have been separated for greater **efficiency**  
and greater **convenience**.” (Norman Bel  
Geddes, 1939)*



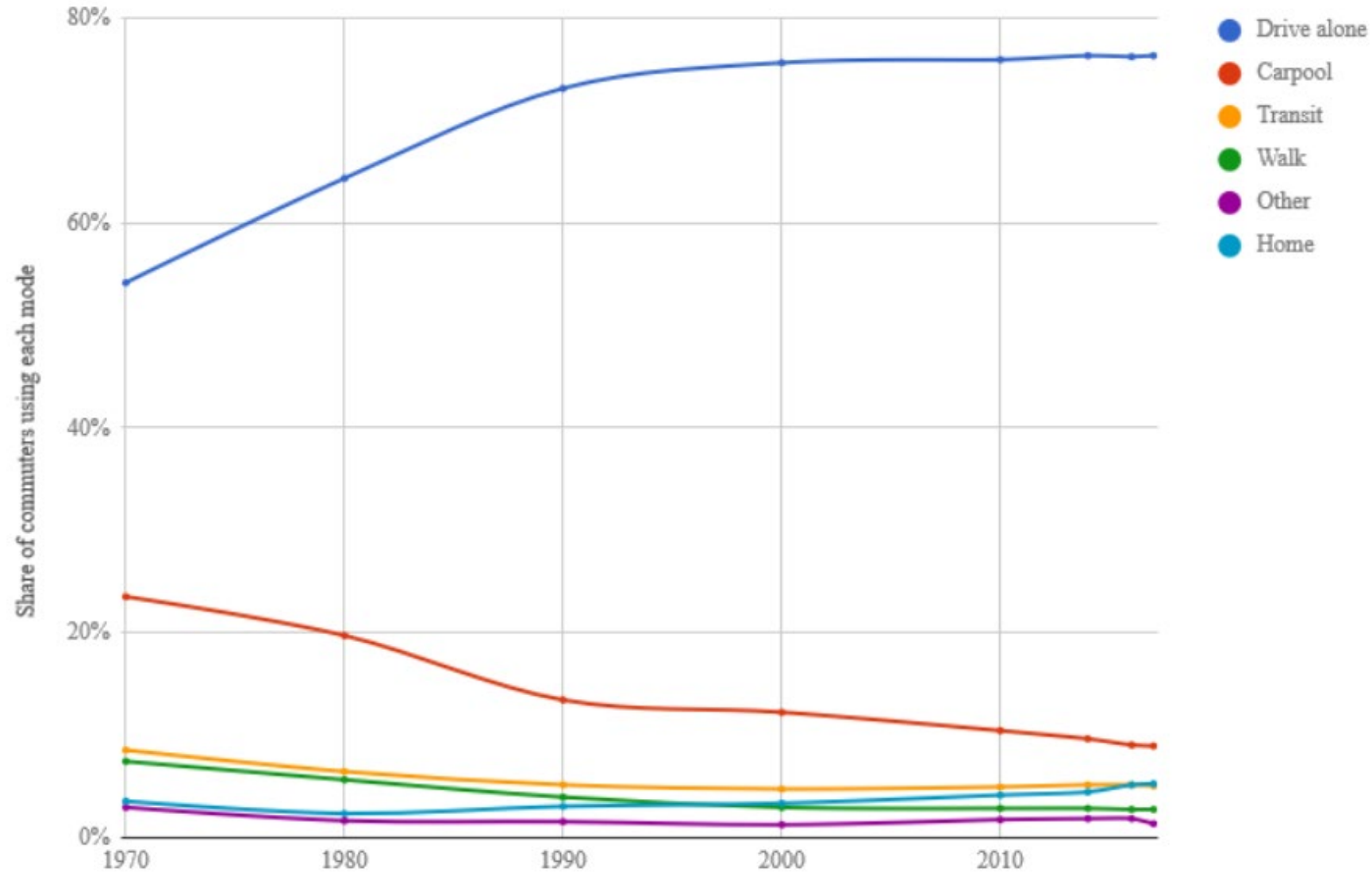




# THE NATIONAL SYSTEM OF INTERSTATE AND DEFENSE HIGHWAYS



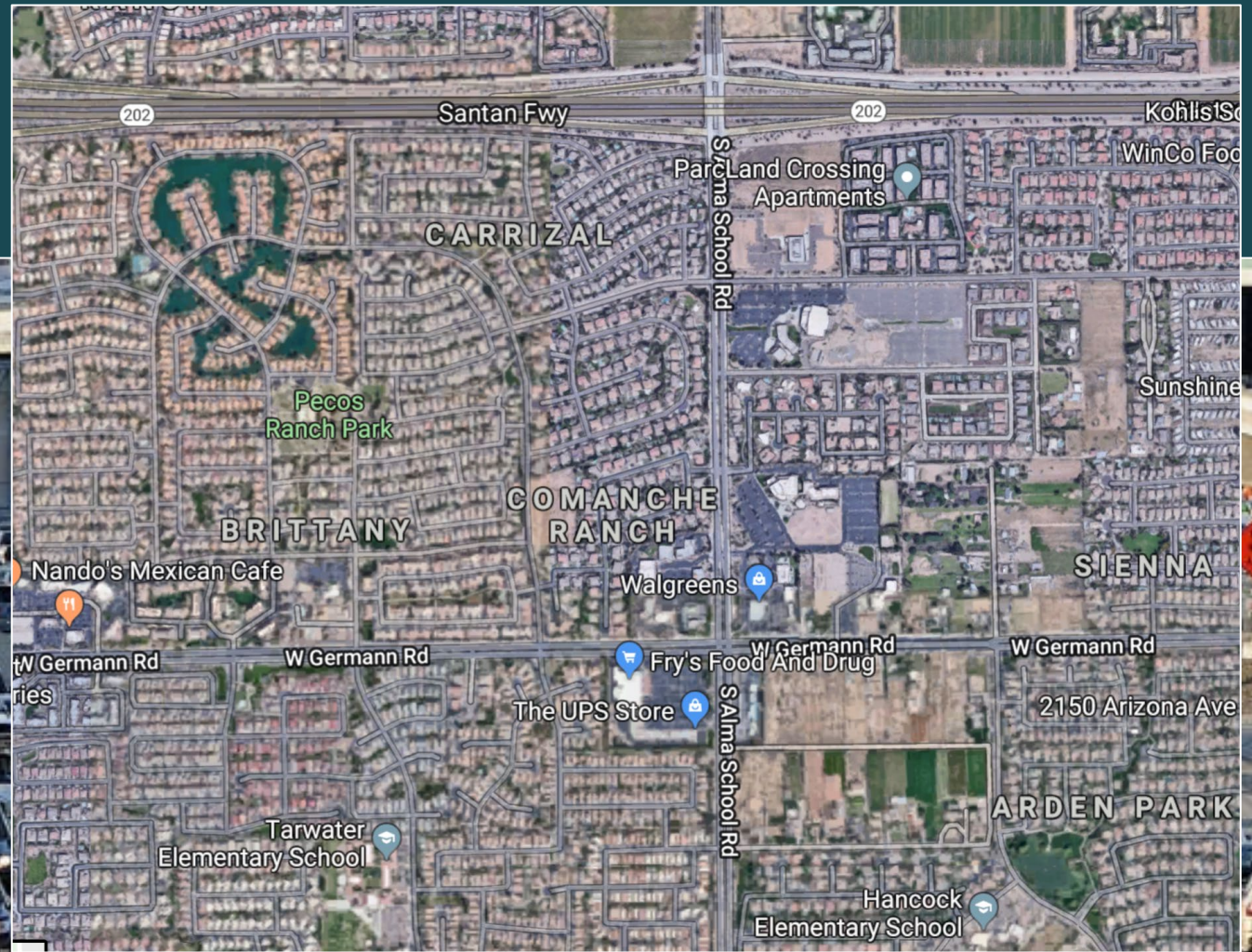
# U.S. COMMUTER MODE SHARE TO WORK, 1970-2017



Created by Yonah Freemark @ The Transport Politic | Source: U.S. Census

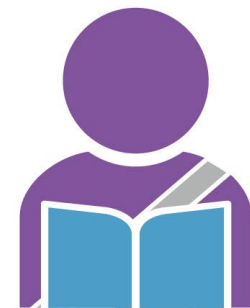
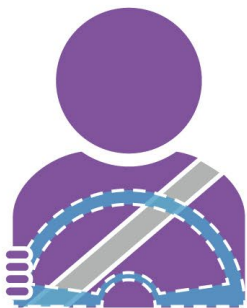
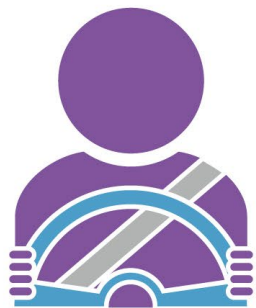


The 1950s through 1990s is marked by geographic dispersion of housing and employment and the provision of **automobile** infrastructure to serve it.





# LEVELS OF DRIVING AUTOMATION



0

1

2

3

4

5

## CONDITIONAL AUTOMATION

Environmental detection capabilities. The vehicle can perform most driving tasks, but human override is still required.

## HIGH AUTOMATION

The vehicle performs all driving tasks under specific circumstances. Geofencing is required. Human override is still an option.

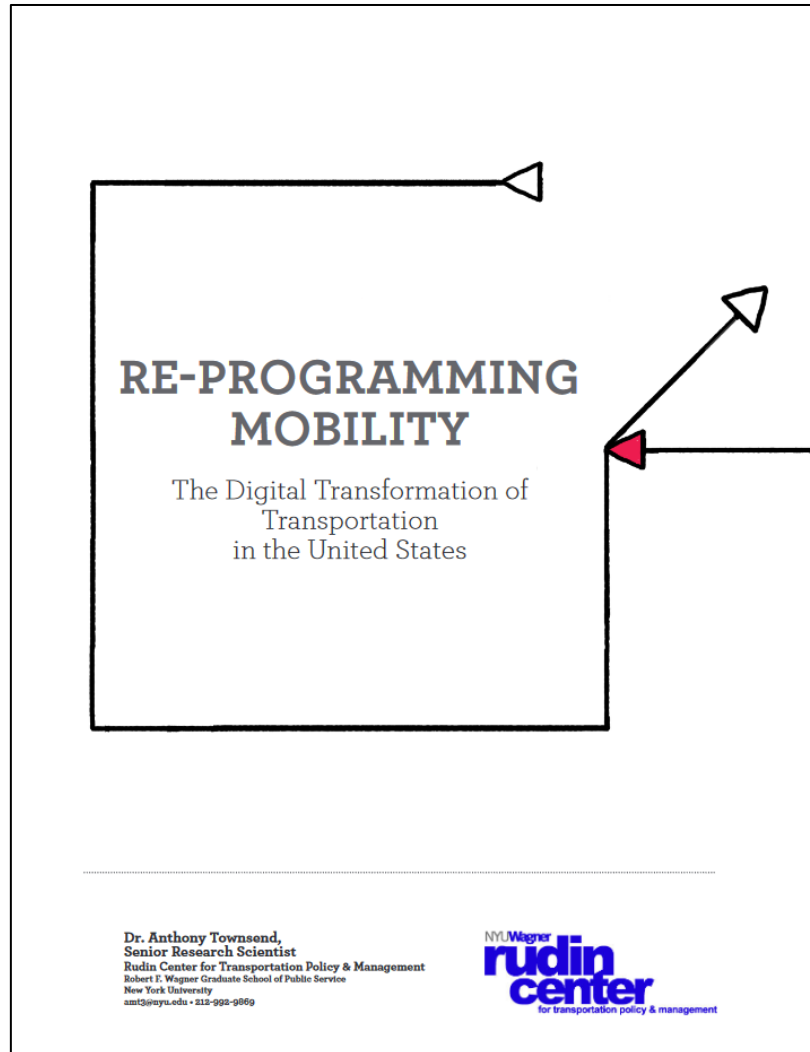
## FULL AUTOMATION

The vehicle performs all driving tasks under all conditions. Zero human attention or interaction is required.

THE AUTOMATED SYSTEM MONITORS THE DRIVING ENVIRONMENT

|                                                                     | Level 0                       | Level 1                               | Level 2                                                                  | Level 3                             | Level 4                       | Level 5                               |
|---------------------------------------------------------------------|-------------------------------|---------------------------------------|--------------------------------------------------------------------------|-------------------------------------|-------------------------------|---------------------------------------|
| <b>Level of Automation</b>                                          | None                          | Driver Assistance                     | Partial                                                                  | Conditional                         | High                          | Full                                  |
| <b>Requires human driver?</b>                                       | Yes                           | Yes                                   | Yes                                                                      | Yes                                 | No                            | No                                    |
| <b>Control speed?</b>                                               | No                            | Yes, but not at same time as steering | Yes, under certain conditions                                            | Yes, under certain conditions       | Yes, under certain conditions | Yes                                   |
| <b>Control steering?</b>                                            | No                            | Yes, under certain conditions         | Yes, under certain conditions                                            | Yes, under certain conditions       | Yes, under certain conditions | Yes                                   |
| <b>Aware of driving environment?</b>                                | No                            | No                                    | No                                                                       | No                                  | No                            | Yes                                   |
| <b>Actions in conditions it cannot drive in (e.g., bad weather)</b> | N/A – Human driver in control | N/A – Human driver in control         | Human driver must decide conditions are not appropriate and take control | Alerts human driver to take control | Implements safety procedures  | N/A – Able to drive in all conditions |

# RE-PROGRAMMING MOBILITY READING



1. What implications do emerging autonomous technologies have on sustainable urban transportation (including public and environmental health) or in what ways could autonomous passenger transportation technologies advance/hinder sustainable forms of urban development?
2. What policy challenges does the advancement of autonomous technologies in passenger transportation systems present?
3. How might transportation planners address the unintended consequences of technology and plan for resilience, happiness and health?



*Transportation touches every aspect of where we live, work, play, and go to school, as well as the physical and natural world. Transportation also plays a pivotal role in shaping human interaction, economic mobility, and sustainability. (Bullard, 2003)*

*Adequate transportation is a precondition for participating in life-enhancing opportunities such as employment, education, health care, and social networks. (Martens and Golub, 2018)*

*Transportation is a basic right, a primary good/service that should be distributed fairly, equitably.*

*The capabilities approach proceeds in three stages:*

- 1. What are the capabilities and functions we want to achieve?*
- 2. What are the minimum conditions necessary to achieve each of these functions and capabilities that enable living a life with dignity by individuals within a region, primarily the most disadvantaged?*
- 3. At the third stage, the approach demands that we design a project that will optimally achieve the approach's goals.*

Hananel, Ravit, and Joseph Berechman. "Justice and Transportation Decision-Making: The Capabilities Approach." *Transport Policy* 49 (July 2016): 78–85.

“Operationalizing and computing geographical accessibility measures depend on a set of four parameters, namely (1) definition of residential areas; (2) a method of aggregation; (3) a measure of accessibility; and (4) a type of distance.”

**Table 1: Approaches for conceptualizing and measuring the geographical accessibility of services and facilities for residential areas**

| Conceptualization                                   | Accessibility measures                                                             |
|-----------------------------------------------------|------------------------------------------------------------------------------------|
| Immediate proximity                                 | The distance between a location and the closest facility                           |
| Availability within one area unit                   | The number of facilities contained within a given unit (for example, census tract) |
| Availability provided by the immediate surroundings | The number of facilities within a given distance from a point of origin            |
| Average cost to reach all destinations              | The average distance between a location and all facilities                         |
| Average cost to reach diversity                     | The average distance between a location and <i>n</i> facilities                    |

Adapted from Talen [41] and Apparicio et al. [32].

Apparicio, Philippe, Mohamed Abdelmajid, Mylène Riva, and Richard Shearmur. 2008. “Comparing Alternative Approaches to Measuring the Geographical Accessibility of Urban Health Services: Distance Types and Aggregation-Error Issues.” *International Journal of Health Geographics* 7 (1): 1–14.

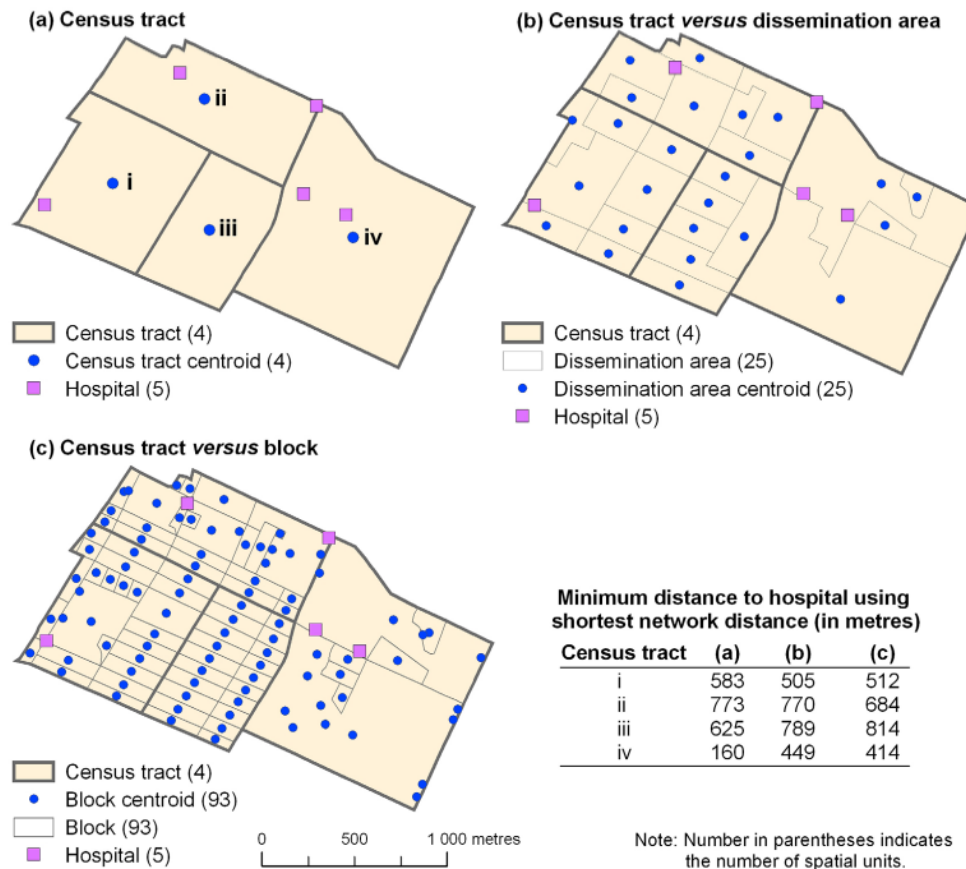
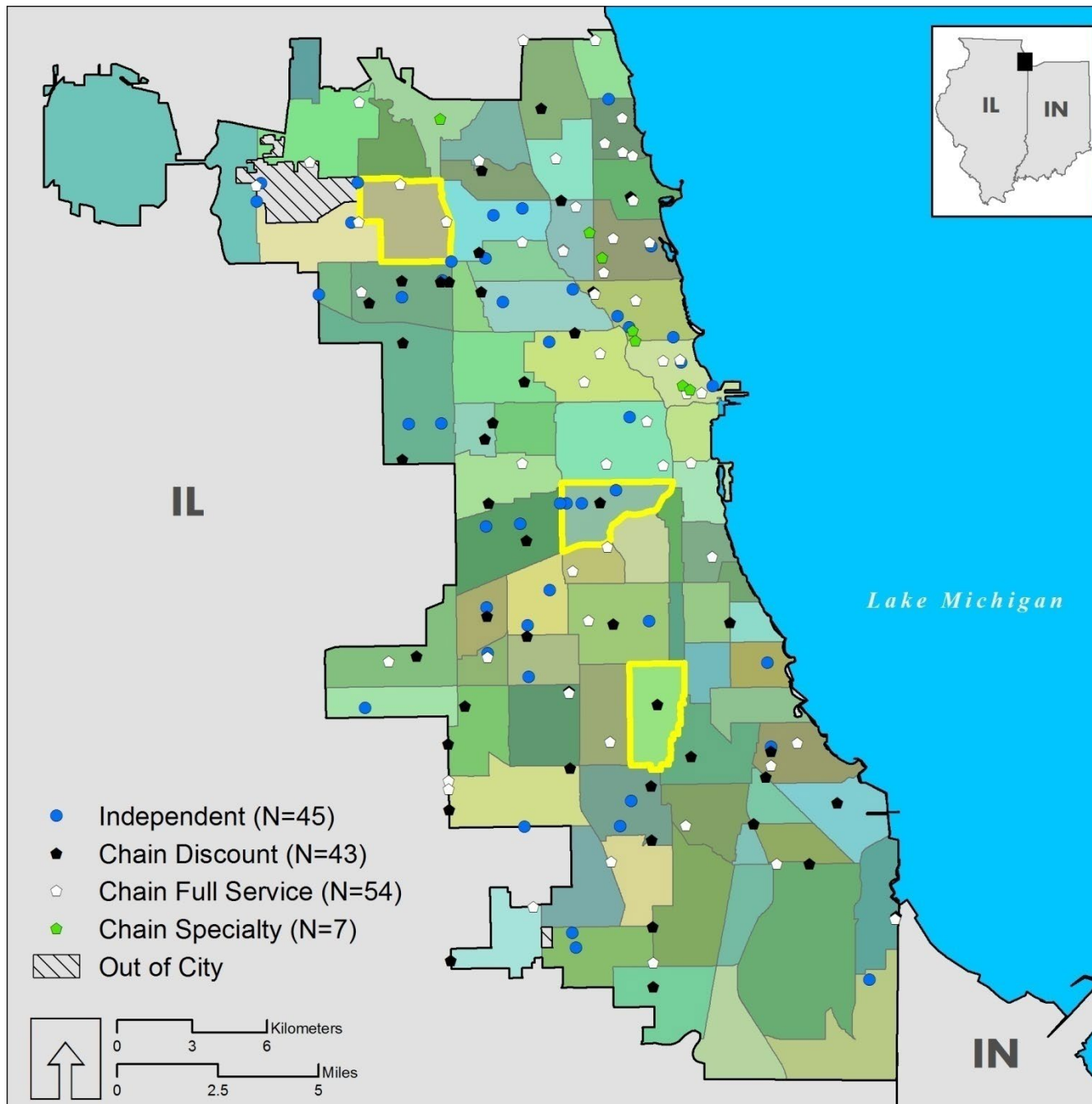


Figure 1  
Choosing the spatial unit of reference for calculating distances and error aggregation.



Figure 2  
Several types of distance.

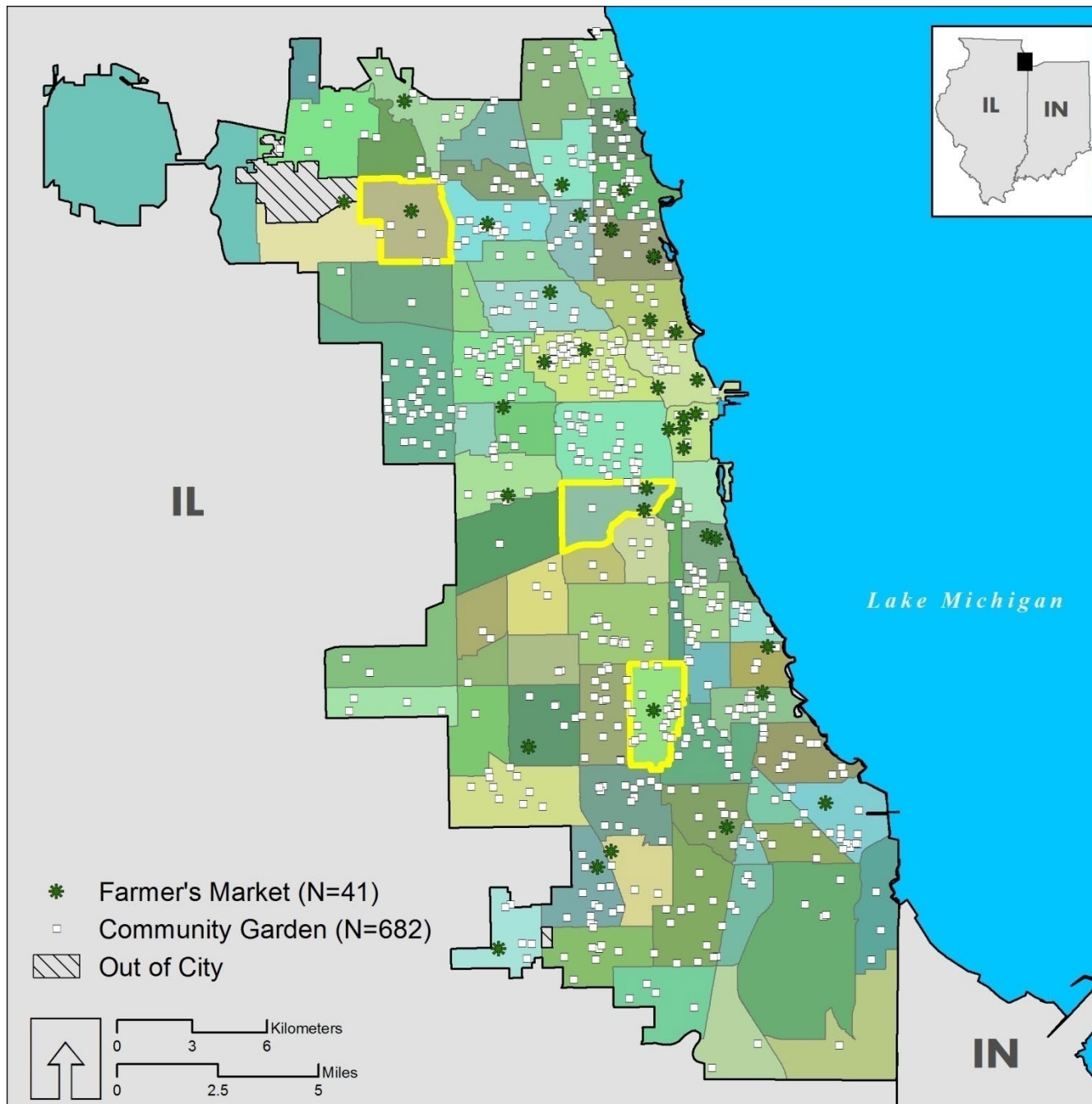
Apparicio, Philippe, Mohamed Abdelmajid, Mylène Riva, and Richard Shearmur. 2008.  
“Comparing Alternative Approaches to Measuring the Geographical Accessibility of Urban Health Services: Distance Types and Aggregation-Error Issues.” *International Journal of Health Geographics* 7 (1): 1–14.



## Distribution of Chain and Independent Grocers

•2007 data of chain and independent grocers made available by Chicago State University



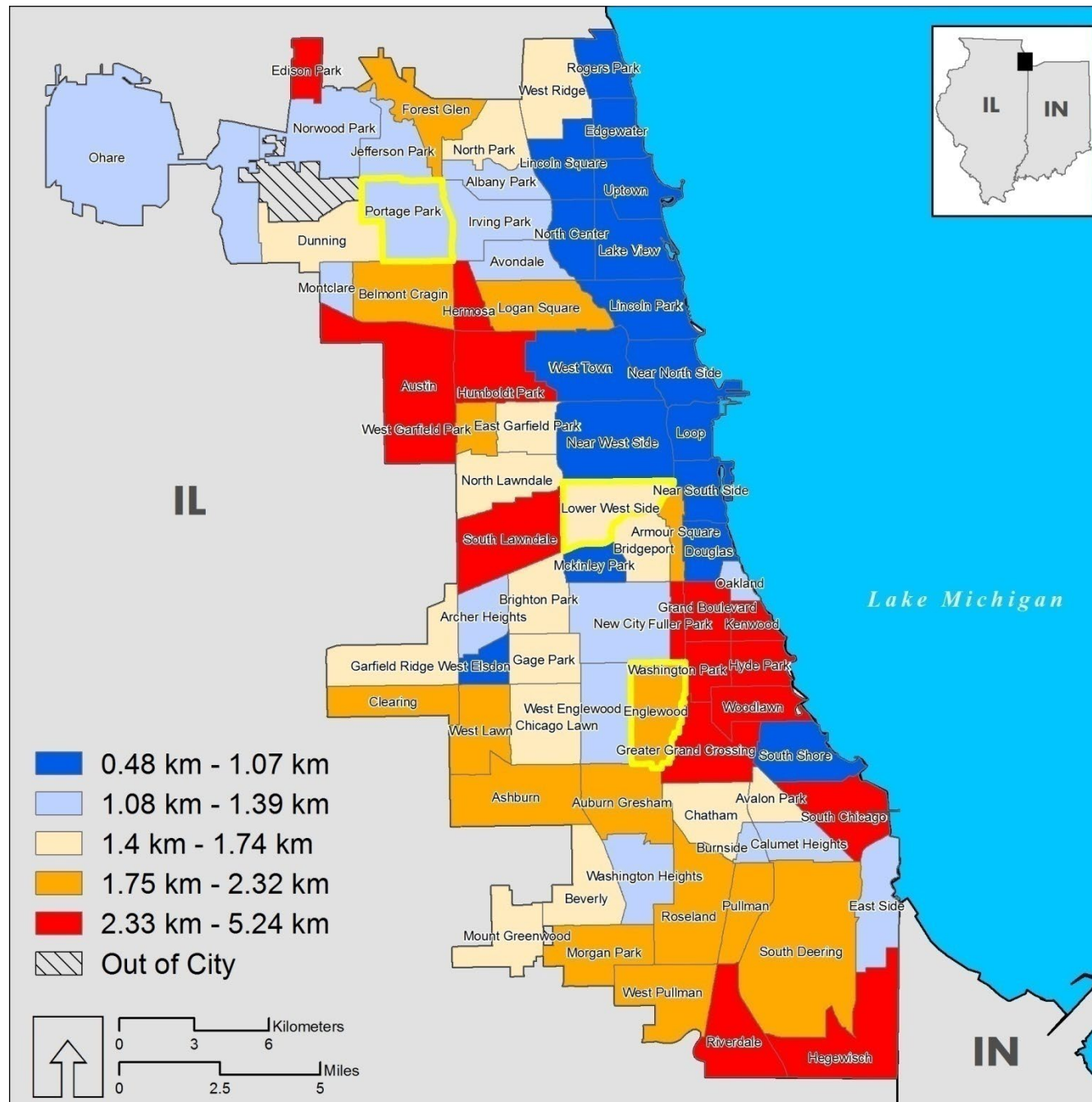


## Distribution of Community Gardens and Farmer's Markets

- 2008/9 Farmer's Markets were compiled from Illinois US Department of Agriculture, on-line listings and field visits;

- Current community gardens dataset compiled (2009) from a range of sources including GreenNet, Chicago Botanic Garden, Neighborspace and



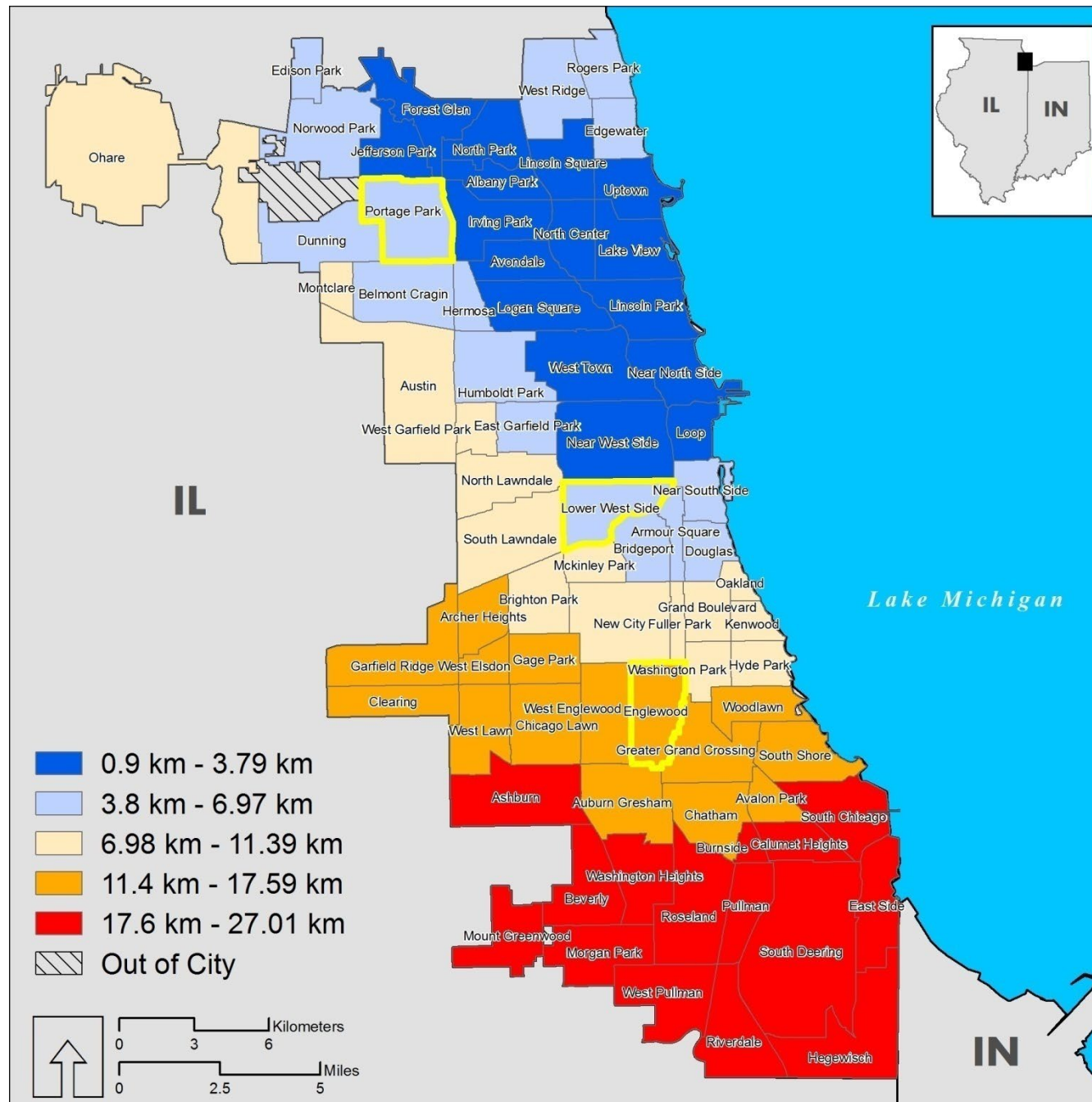


**Average Distance Weighted, Full Service Grocers**

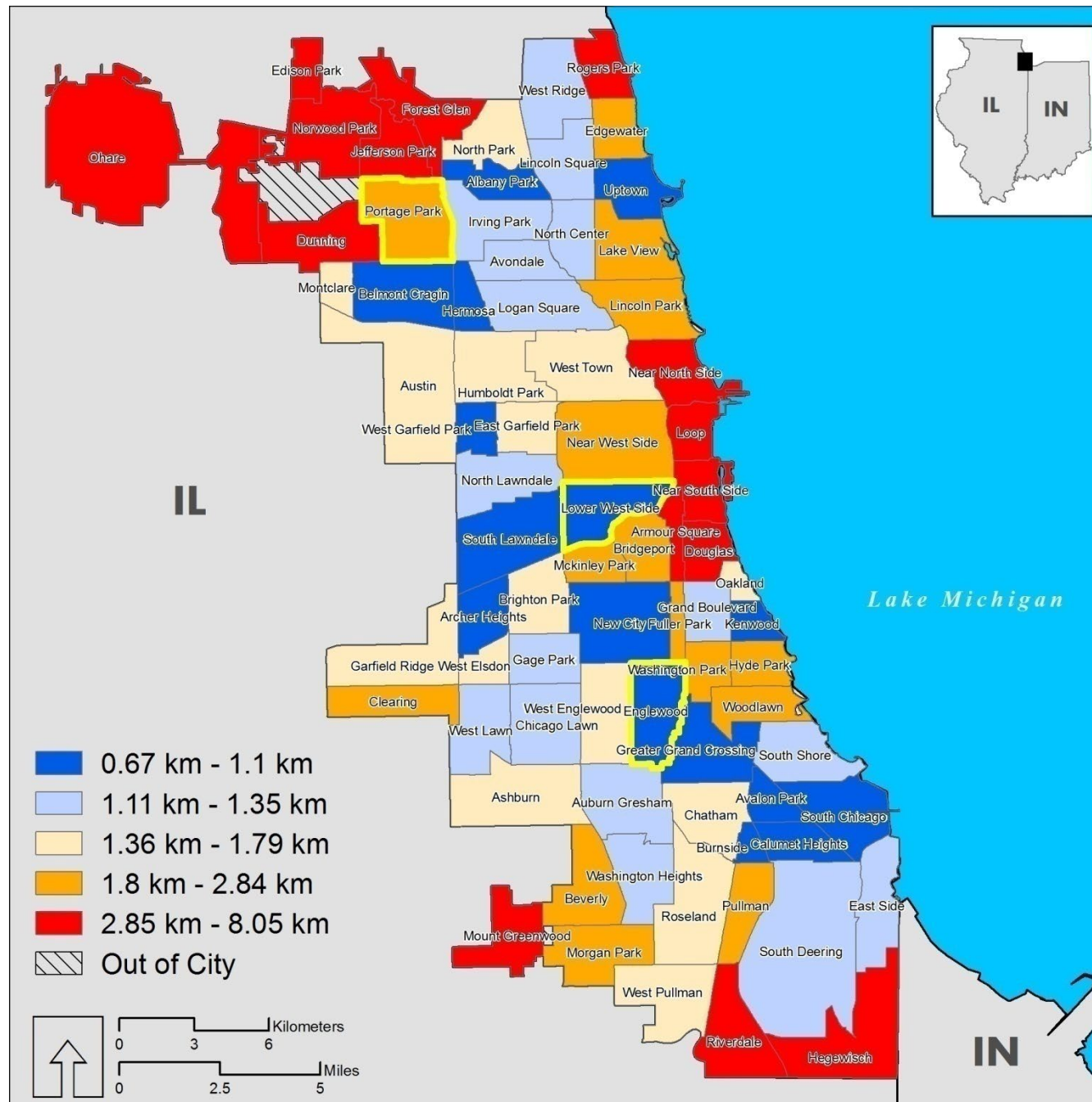
- Average distances were population-weighted based on distances calculated from census blocks (N=18,266) to the nearest food source (in this case, the nearest full service grocer)



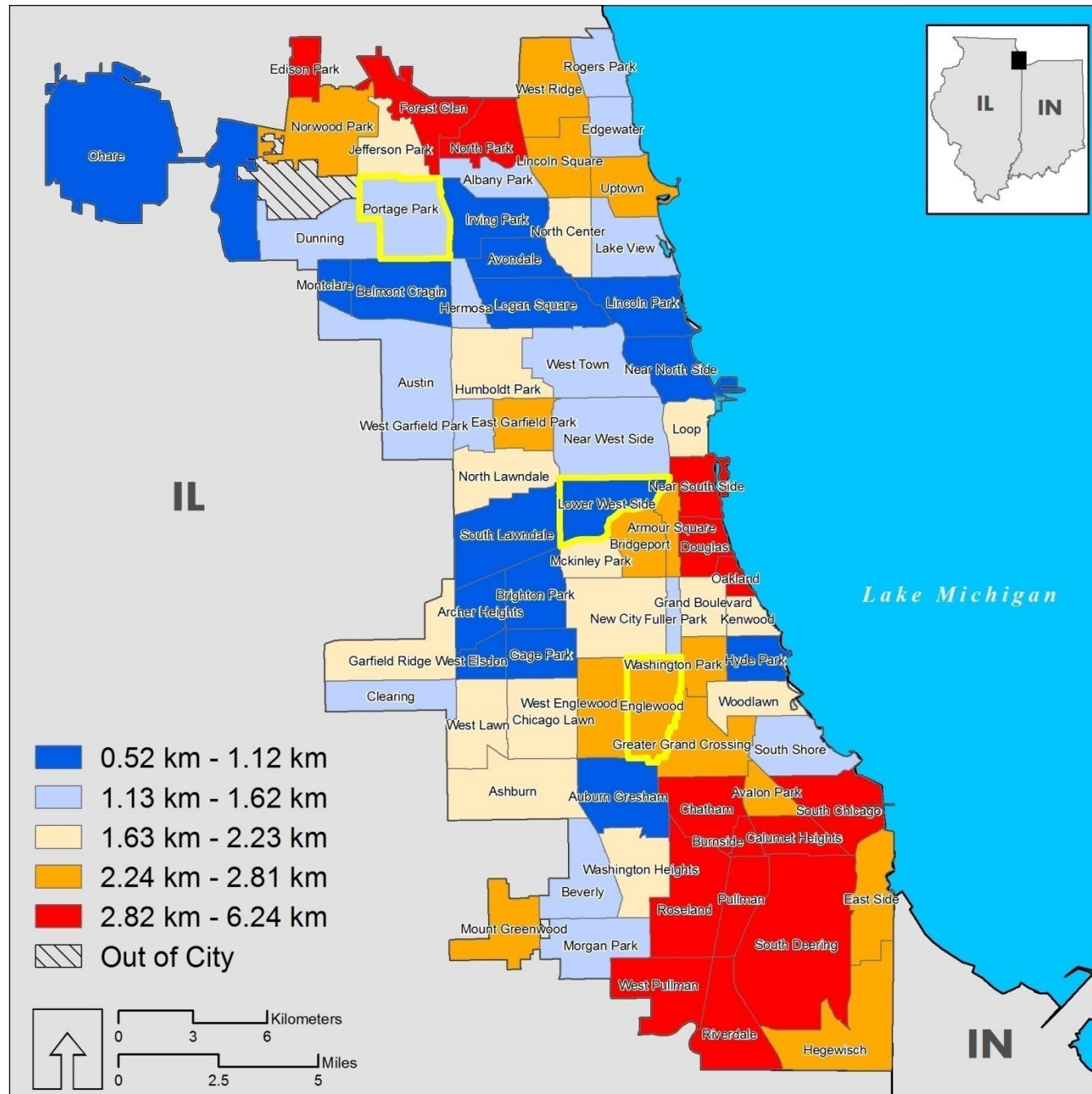
# Average Distance Weighted, Specialty Grocers



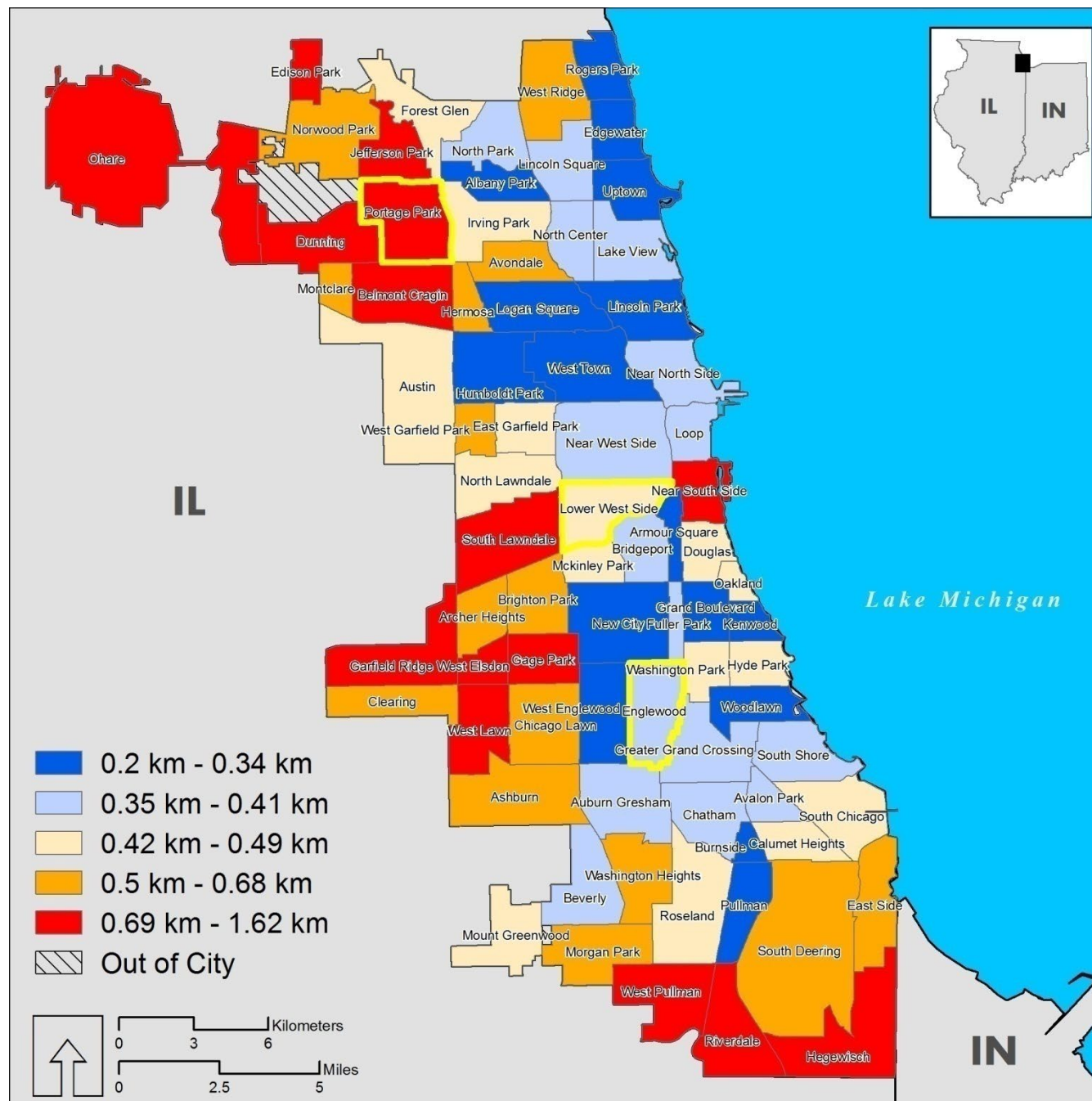
# Average Distance Weighted, Discount Grocers



# Average Distance Weighted, Large Independent Grocers







# Average Distance Weighted, Community Gardens



# Bivariate Spatial Association Results

|                                      | Distance to<br>Specialty<br>Grocer | Distance to<br>Full Service<br>Grocer | Distance to<br>Discount<br>Grocer | Distance to<br>Large<br>Independe<br>nt Grocer | Distance to<br>Farmer's<br>Market | Distance to<br>Community<br>Garden* |
|--------------------------------------|------------------------------------|---------------------------------------|-----------------------------------|------------------------------------------------|-----------------------------------|-------------------------------------|
| % Black or African American<br>Alone | <b><i>0.4232</i></b>               | <b><i>0.2690</i></b>                  | <b><i>-0.2619</i></b>             | <b><i>0.4268</i></b>                           | -0.0788                           | <b><i>-0.2921</i></b>               |
| % Hispanic or Latino                 | -0.1604                            | -0.0476                               | <b><i>-0.2943</i></b>             | <b><i>-0.4257</i></b>                          | 0.1332                            | 0.1581                              |
| % White Alone                        | <b><i>-0.3206</i></b>              | <b><i>-0.2716</i></b>                 | <b><i>0.5668</i></b>              | -0.2144                                        | 0.0571                            | <b><i>0.3137</i></b>                |
| % HH Income > \$50K                  | -0.0525                            | <b><i>-0.2503</i></b>                 | <b><i>0.4674</i></b>              | 0.0079                                         | -0.0665                           | 0.1214                              |
| % Bachelors Degree or Higher         | <b><i>-0.4399</i></b>              | <b><i>-0.3328</i></b>                 | <b><i>0.3012</i></b>              | -0.1619                                        | <b><i>-0.4197</i></b>             | <b><i>-0.2354</i></b>               |
| % Population in Poverty              | 0.0479                             | <b><i>0.3058</i></b>                  | <b><i>-0.2299</i></b>             | <b><i>0.2711</i></b>                           | -0.0539                           | <b><i>-0.3000</i></b>               |
| % Vacant Housing Units               | -0.0705                            | 0.1798                                | -0.0866                           | 0.1409                                         | -0.1684                           | <b><i>-0.2993</i></b>               |

Values in ***bold, italics*** significant at  $p < 0.05$  level

The locations of farmer's markets tend to be race neutral whereas ***proximity*** to community gardens varies positively with percent African American and negatively with percent White.





# TOWARD UNIVERSAL MOBILITY:

Charting a Path to Improve Transportation Accessibility

December 2019



Figure 7. An example of a bus stop with 50% sidewalk network completeness for streets within a half mile

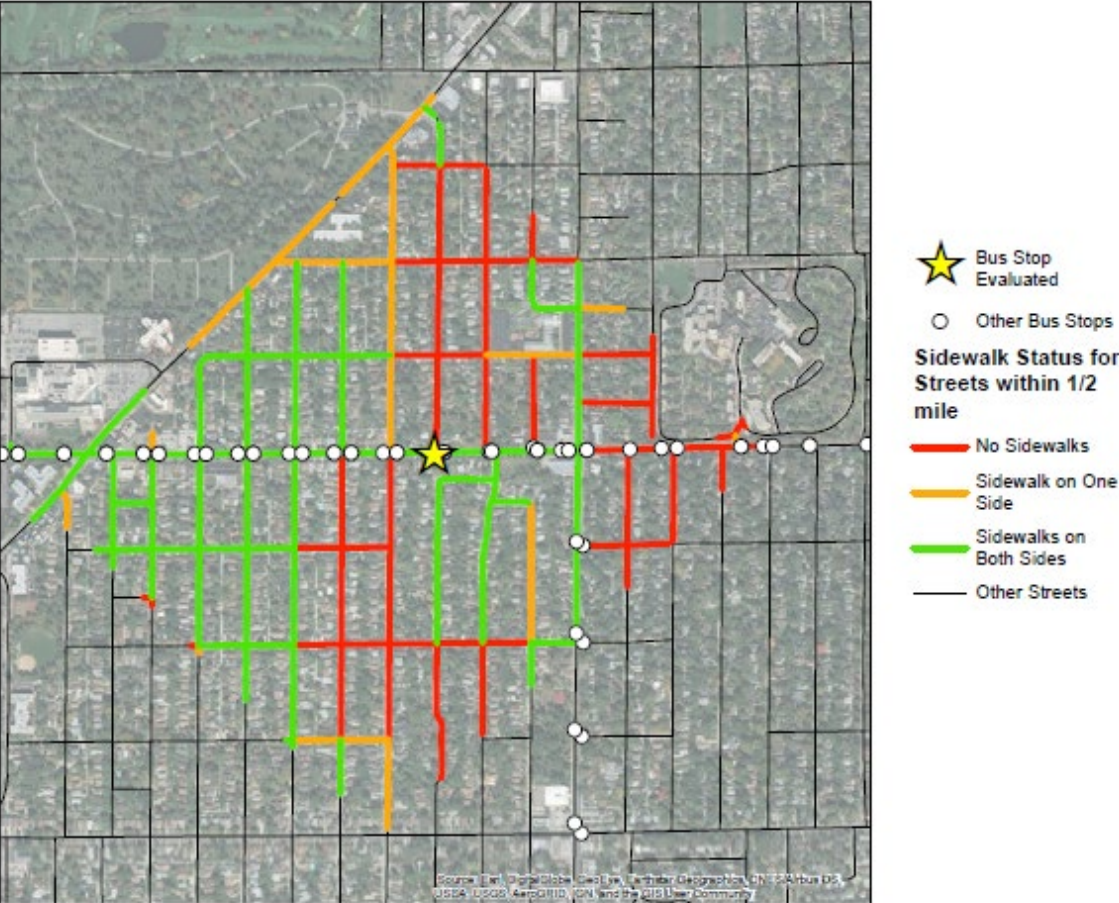


Figure 8. Sidewalk Network Completeness Near Pace Bus Stops

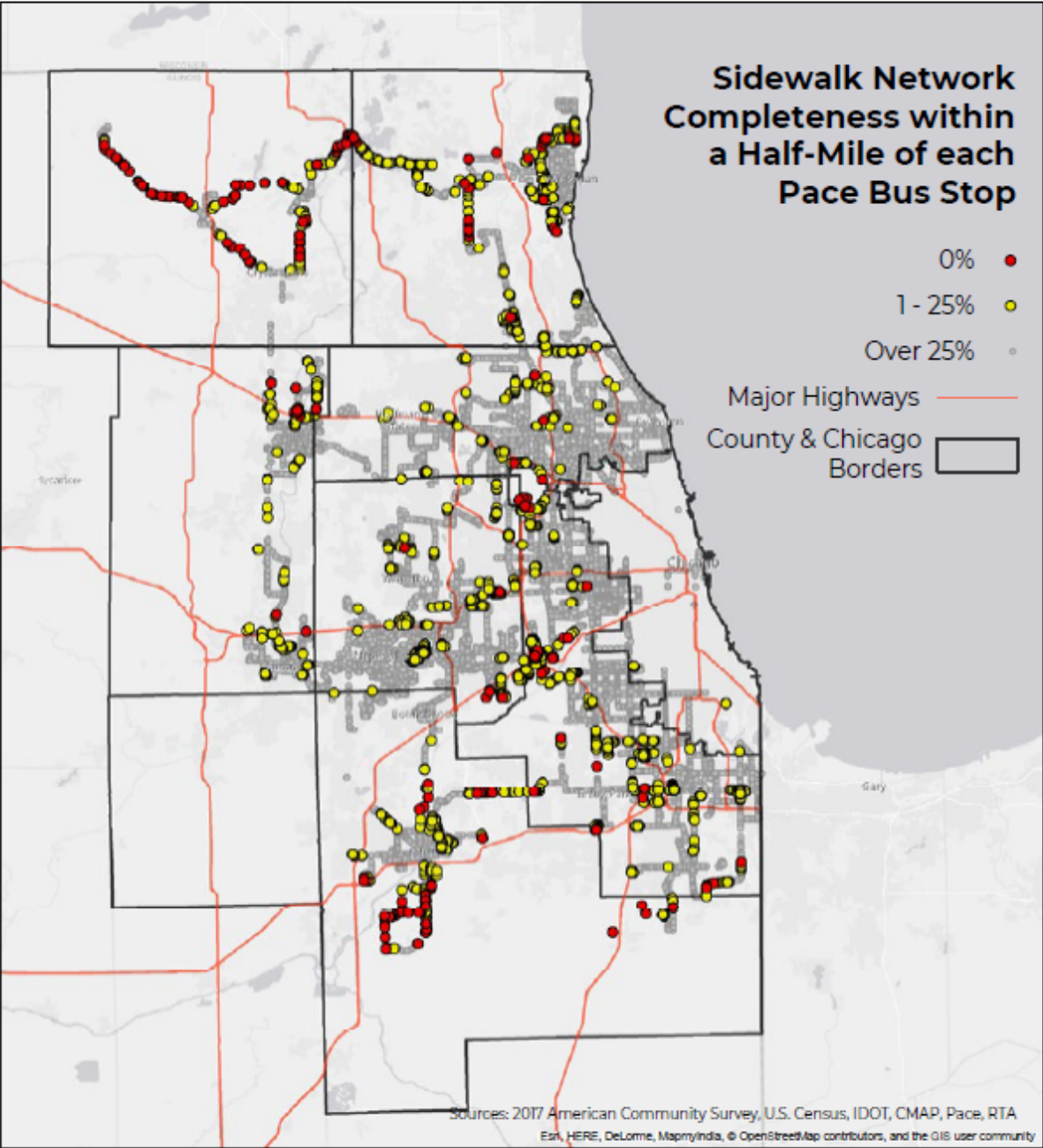


Figure 7. An example of a bus stop with 50% sidewalk network completeness for streets within a half mile

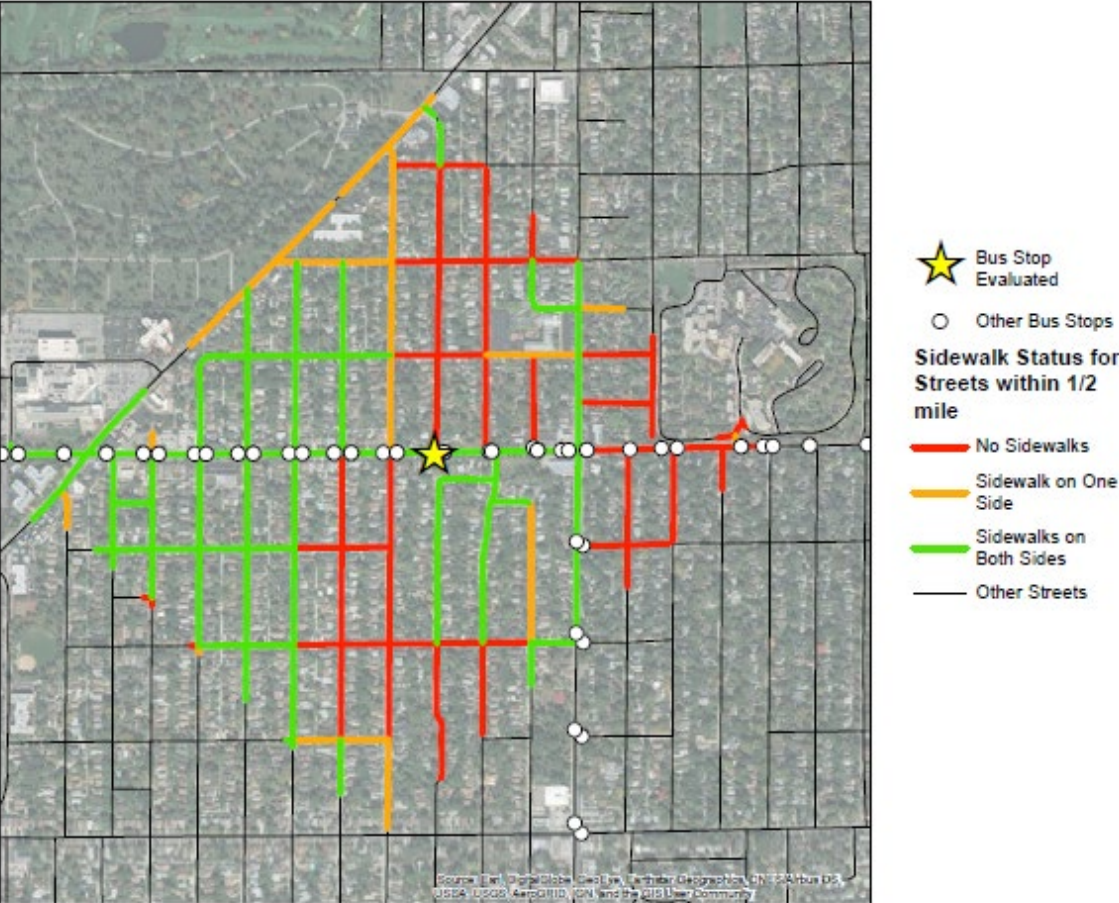
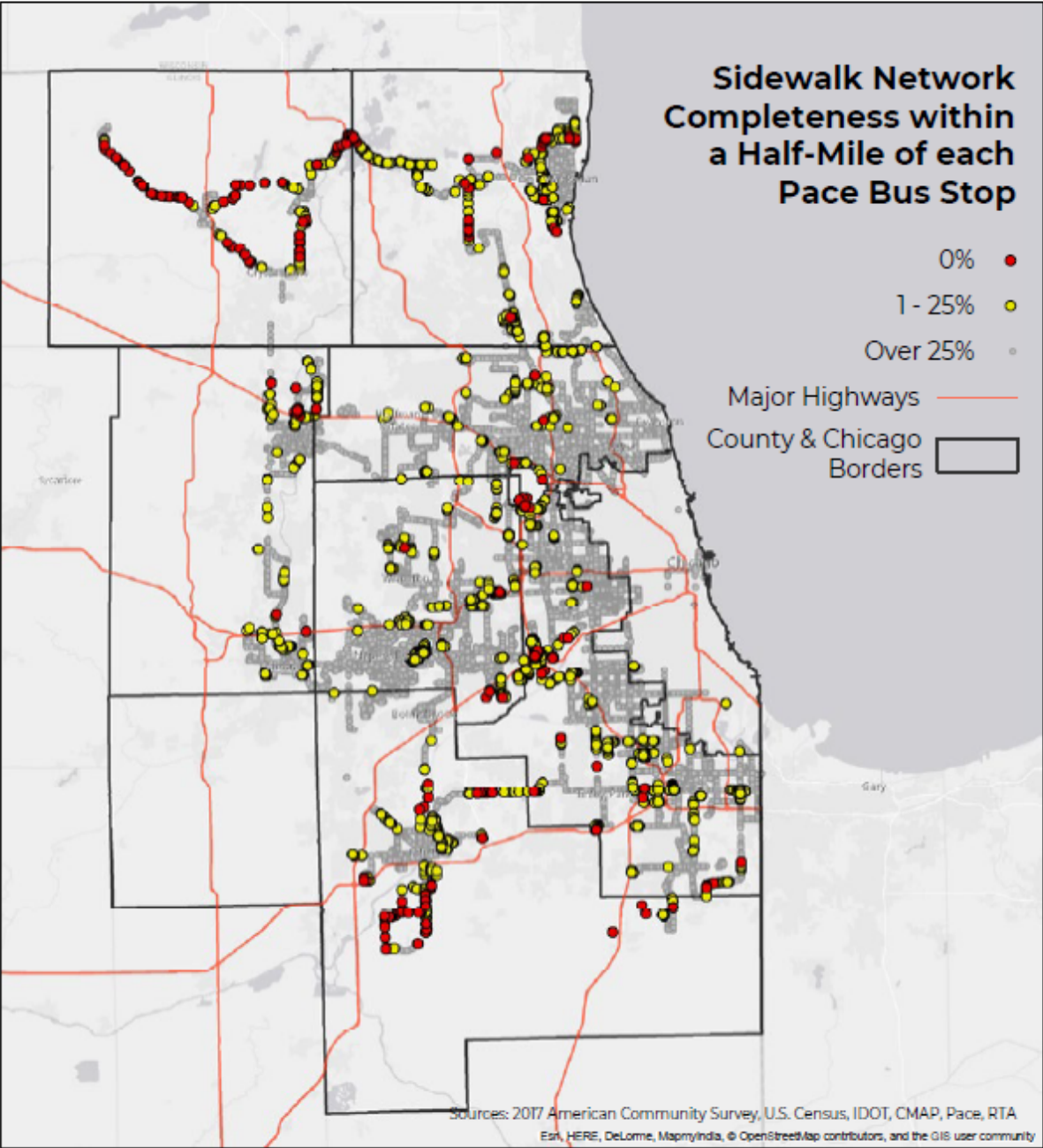
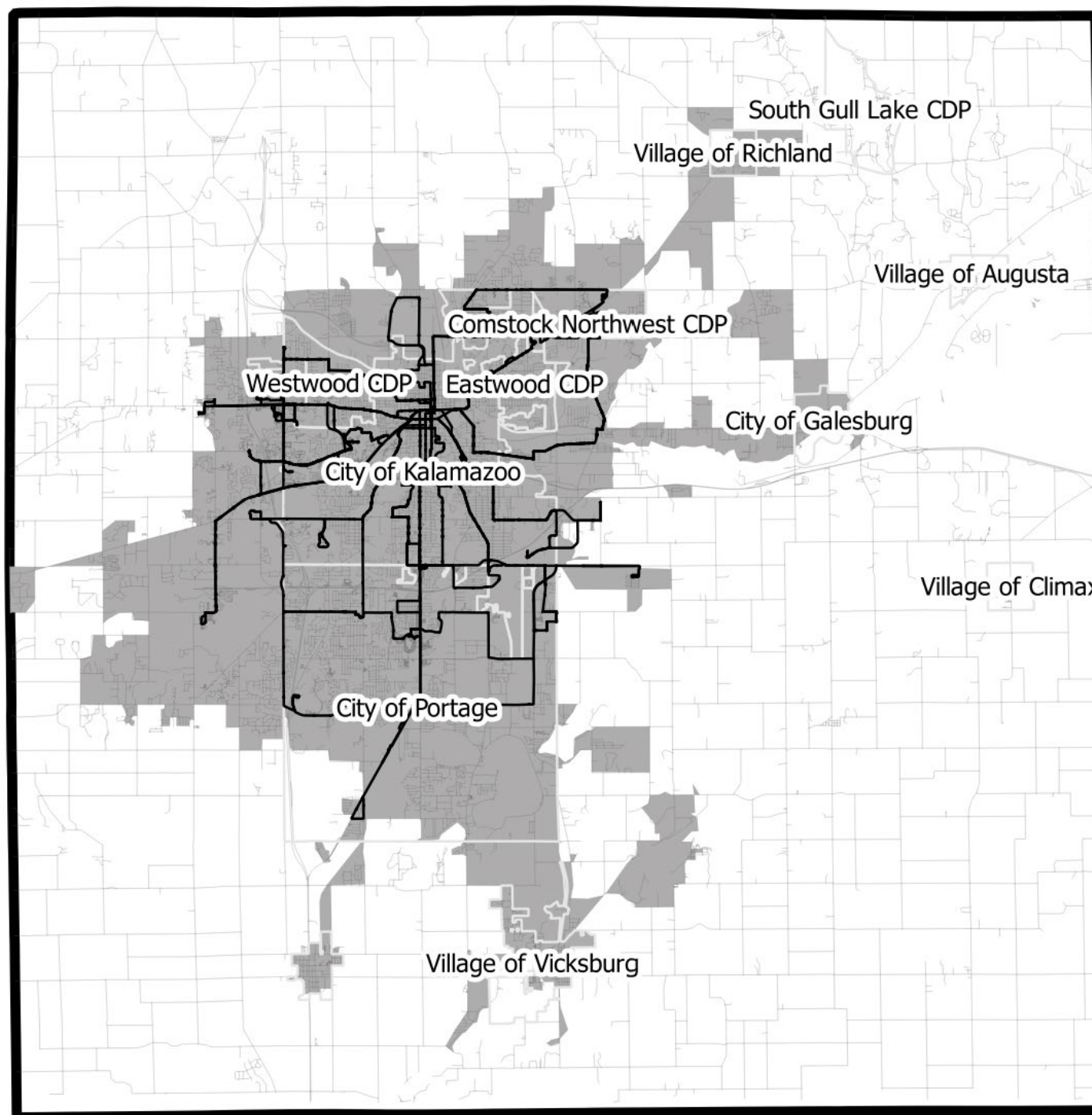
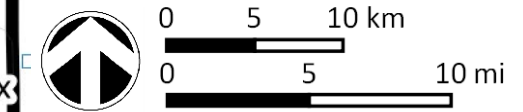


Figure 8. Sidewalk Network Completeness Near Pace Bus Stops

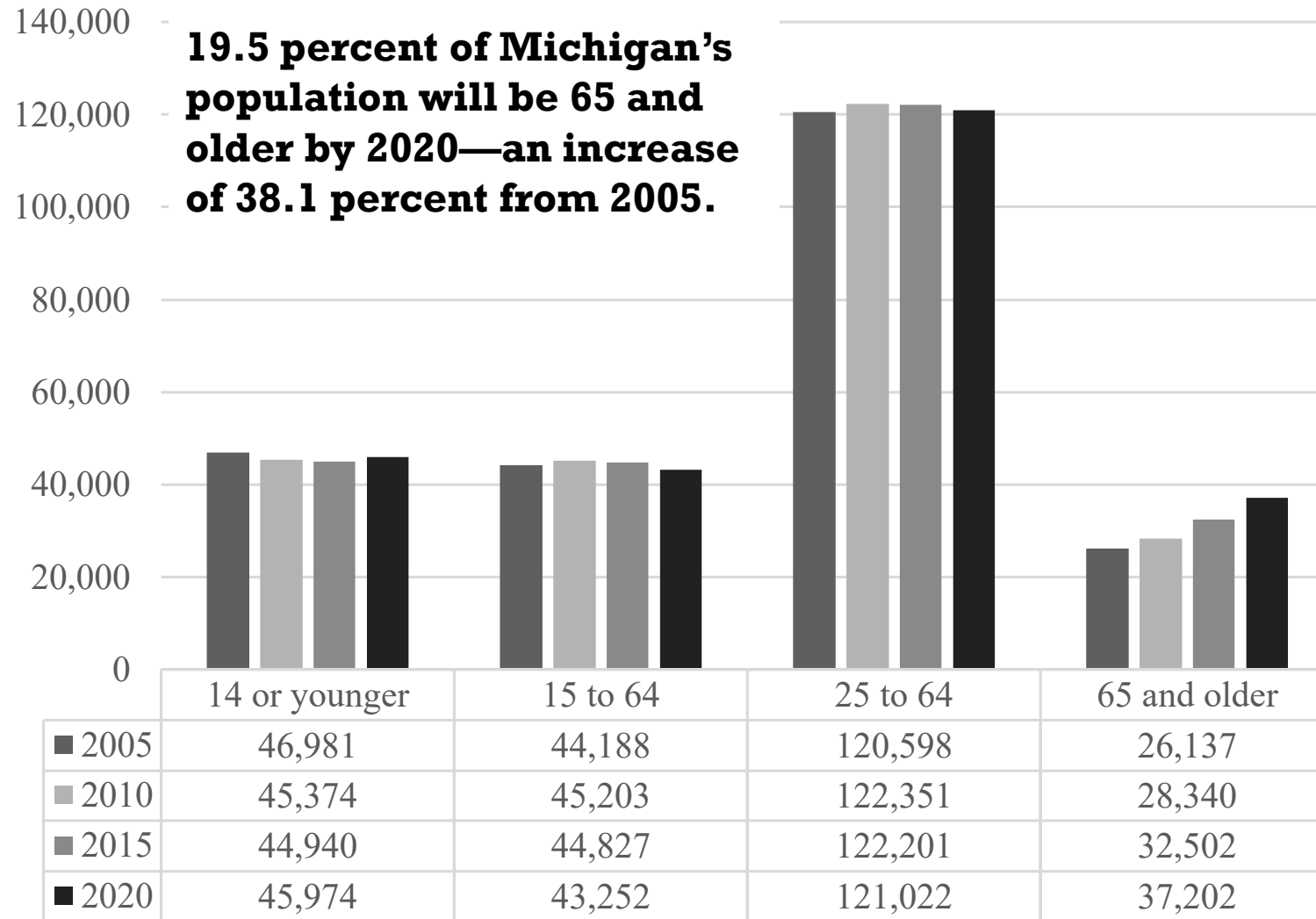


# Kalamazoo County Study Area

-  Townships
-  Places
-  Kalamazoo County
-  Roads



# Kalamazoo County Population Trends & Projection by Age Group (2005-2020)



Source: US Census Bureau ACS, 2015



Table 1. Percent of People in the Chicago Region Injured in a Crash that are 65 or older, 2013-2017

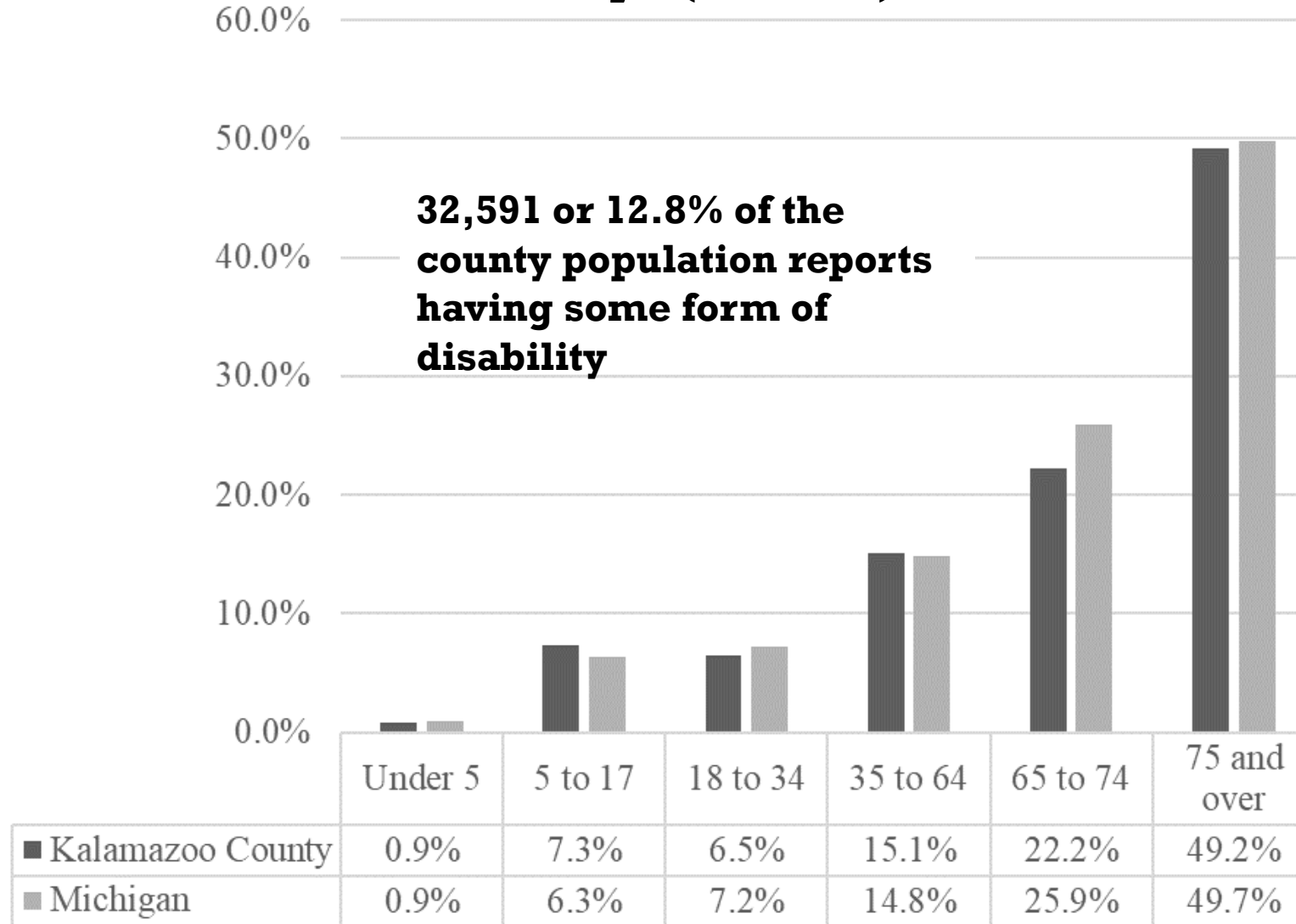
| Severity        | All Crash Victims | Pedestrians Only |
|-----------------|-------------------|------------------|
| No Injury       | 7.7%              | 7.2%             |
| Possible Injury | 8.7%              | 8.2%             |
| Minor Injury    | 8.6%              | 9.8%             |
| Serious Injury  | 9.7%              | 11.3%            |
| Fatal           | 18.2%             | 25.2%            |

Source, Chicago Metropolitan Agency for Planning; data for Cook, Lake, DuPage, Kane, McHenry and Will Counties

Over 25% of pedestrian fatalities were seniors, despite making up only around 12% of the region's population.



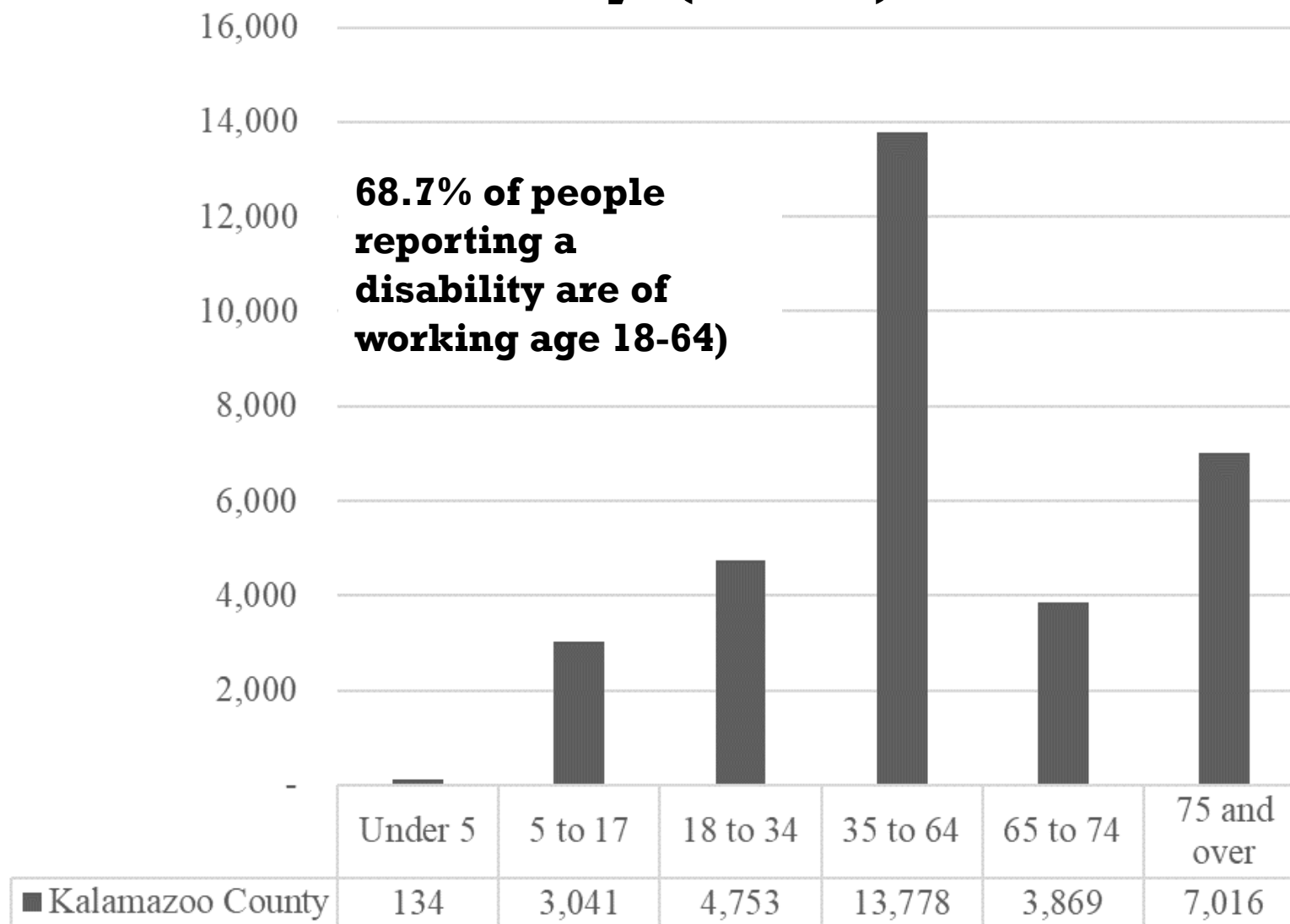
# Disability Status by Age Group Kalamazoo County (2015)



Source: US Census Bureau ACS, 2015



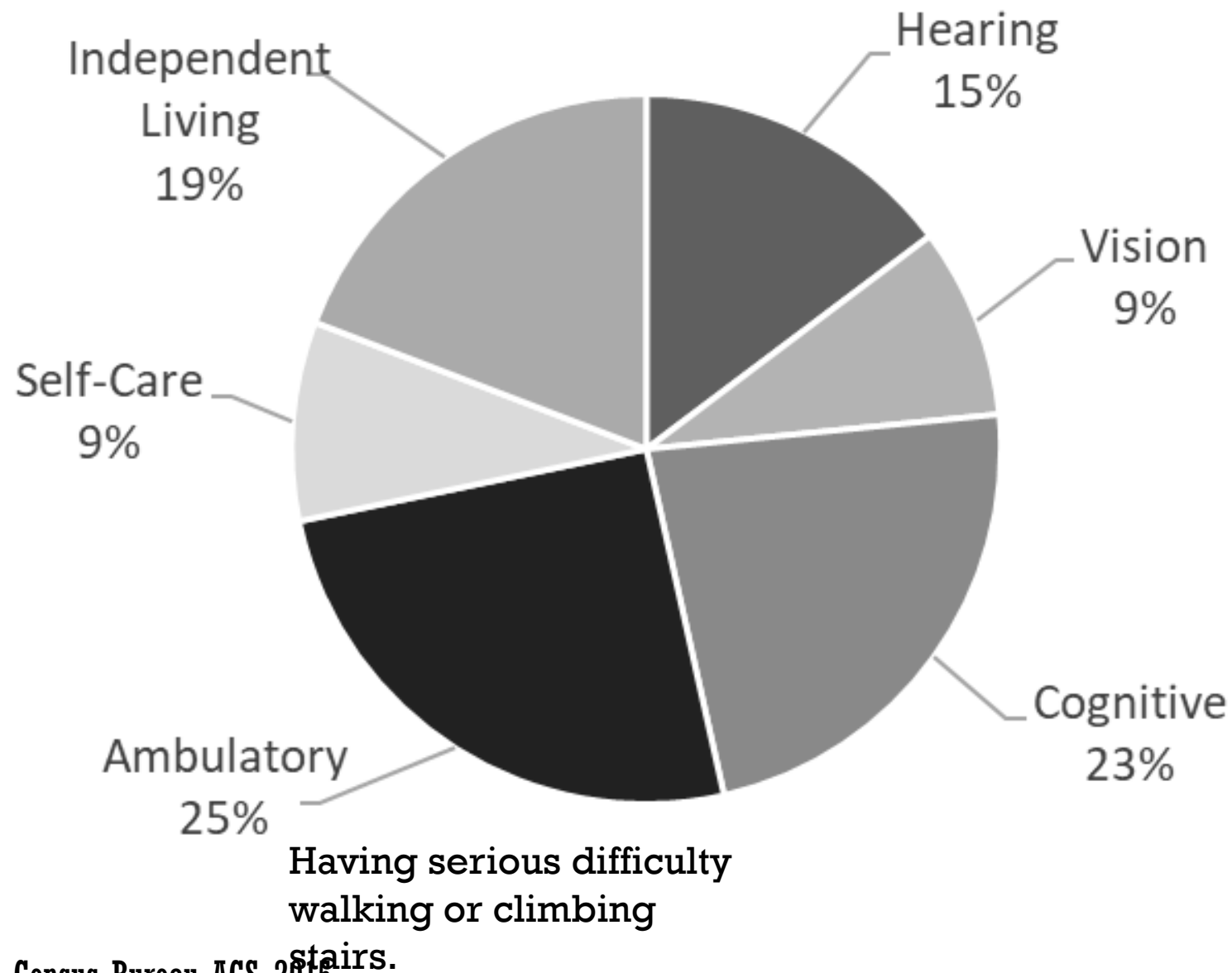
# Disability Status by Age Group Kalamazoo County (2015)



Source: US Census Bureau ACS, 2015



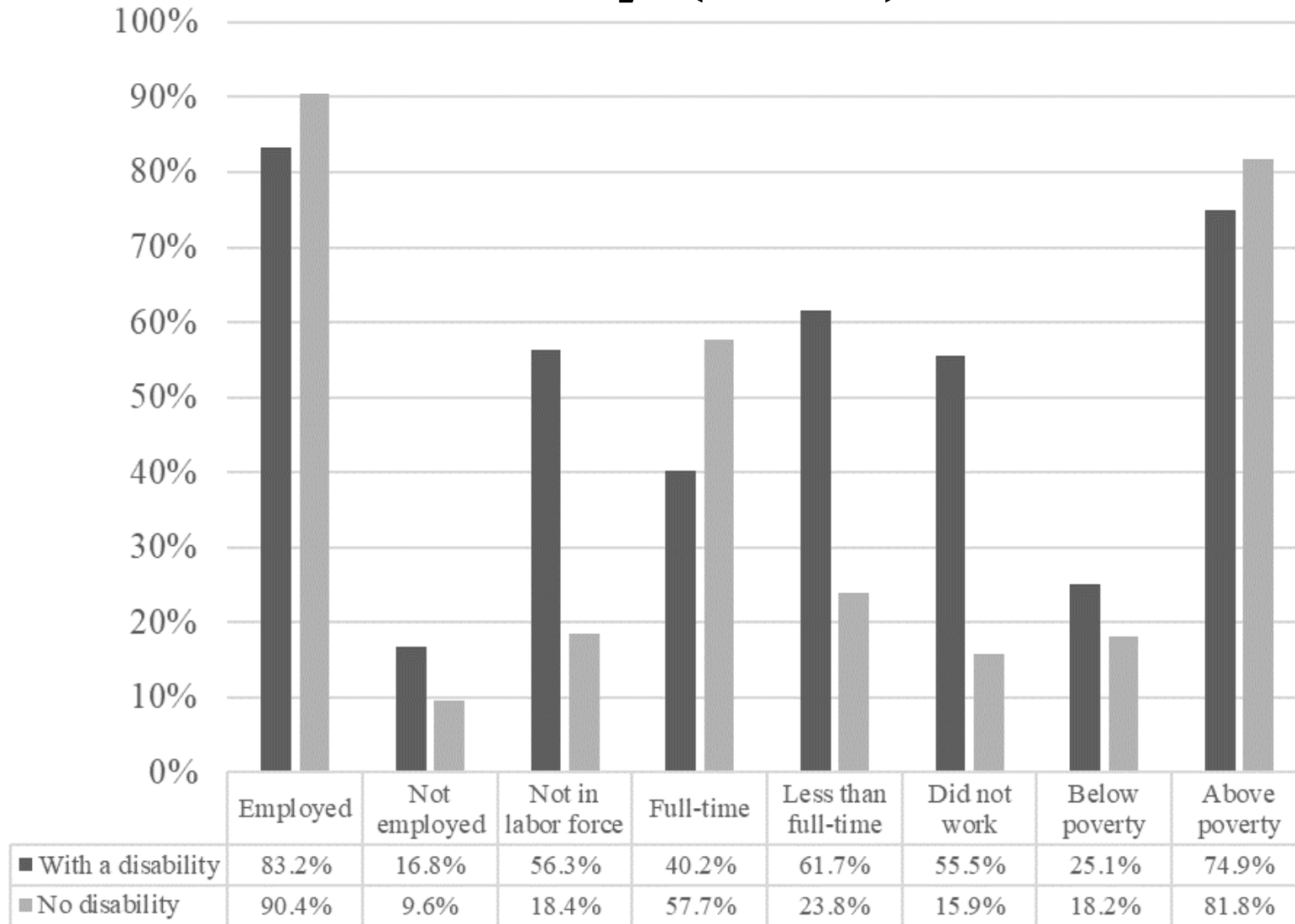
# Disability Category (2015)



Source: US Census Bureau ACS, 2015



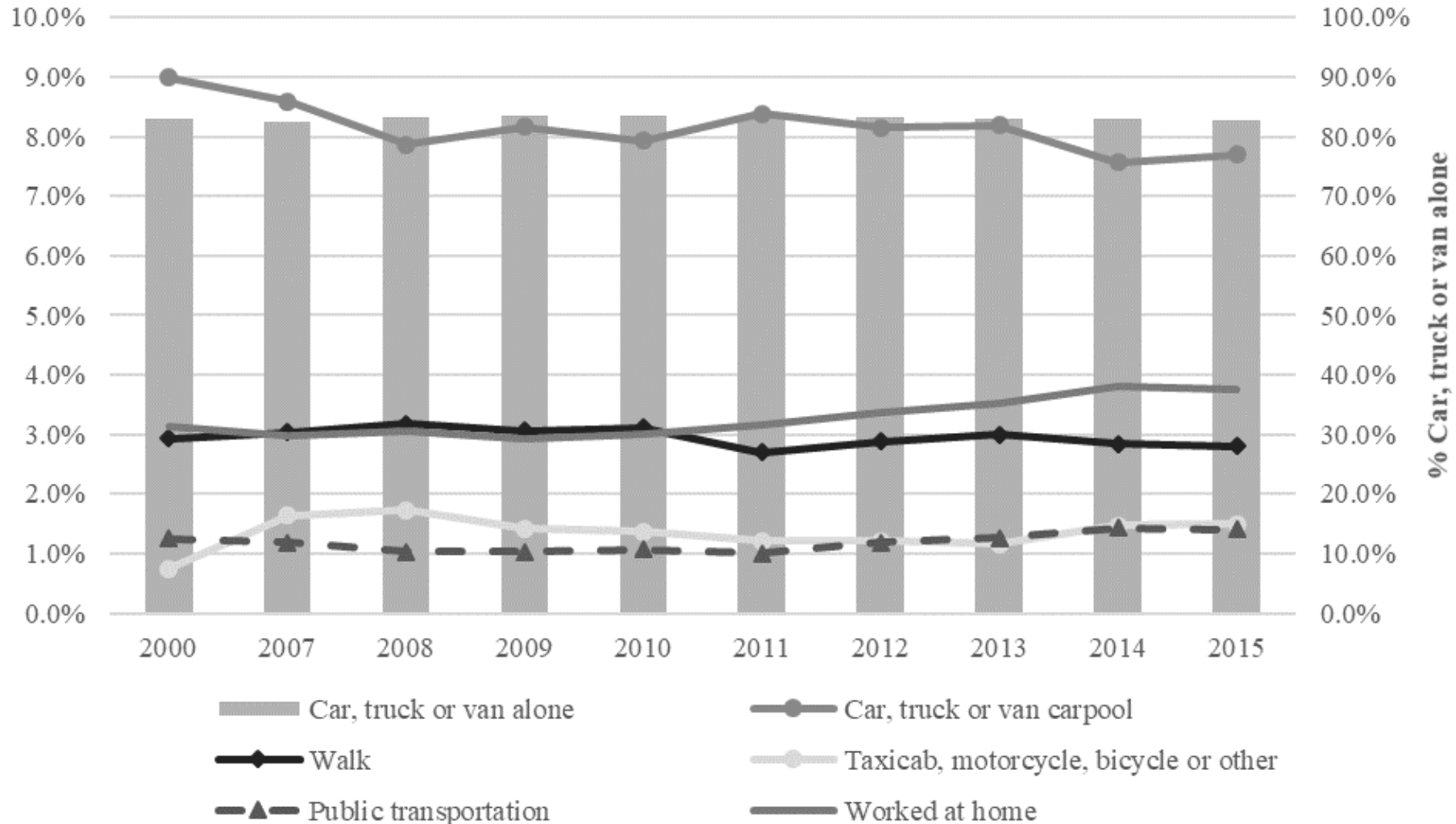
# Employment by Disability Status Kalamazoo County (2015)



Source: US Census Bureau ACS, 2015

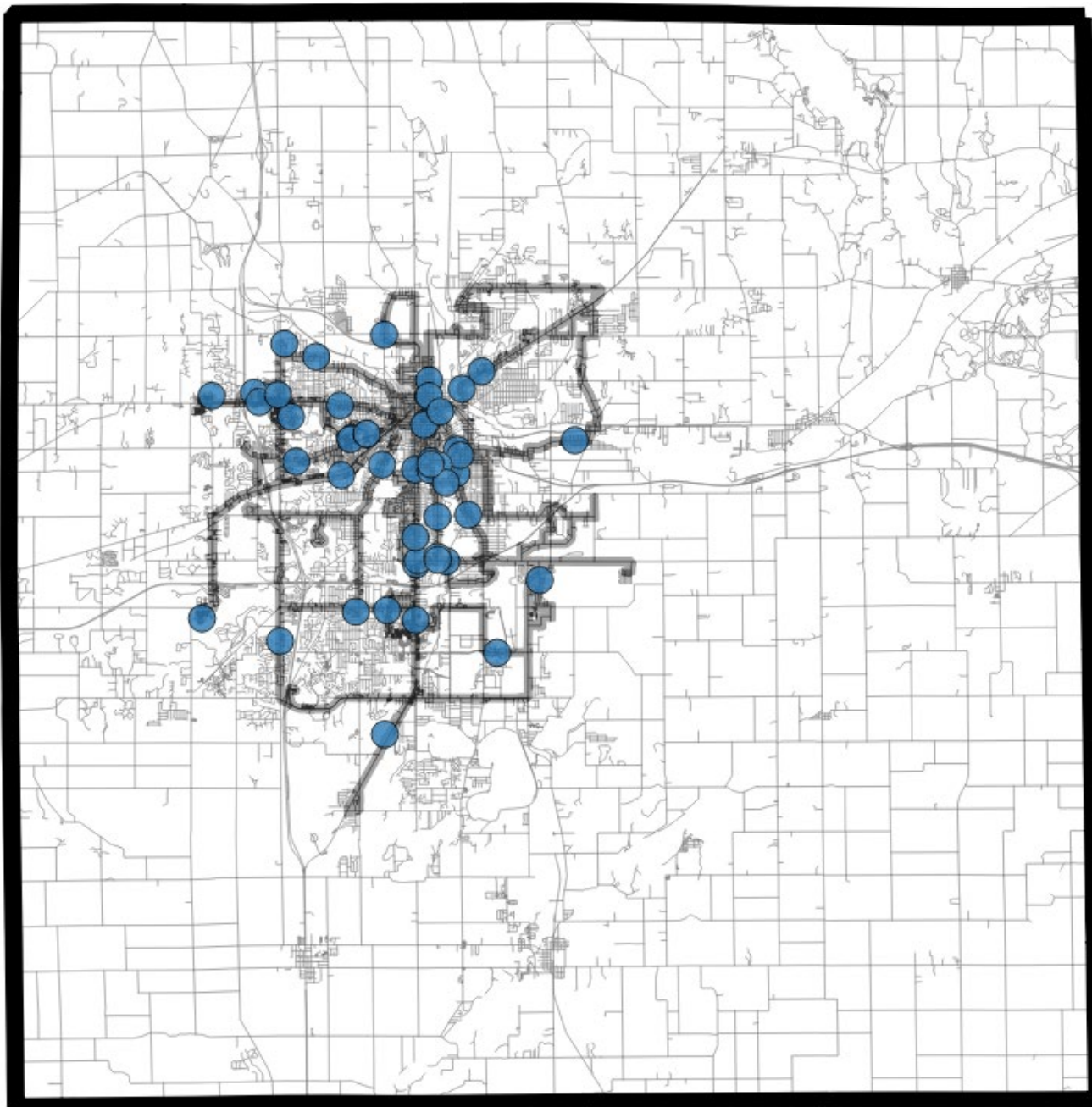


# Work Commute Mode Share, Kalamazoo County (2000-2015)



Source: US Census Bureau ACS, 2000-2015





## Kalamazoo Metro Bus Stops

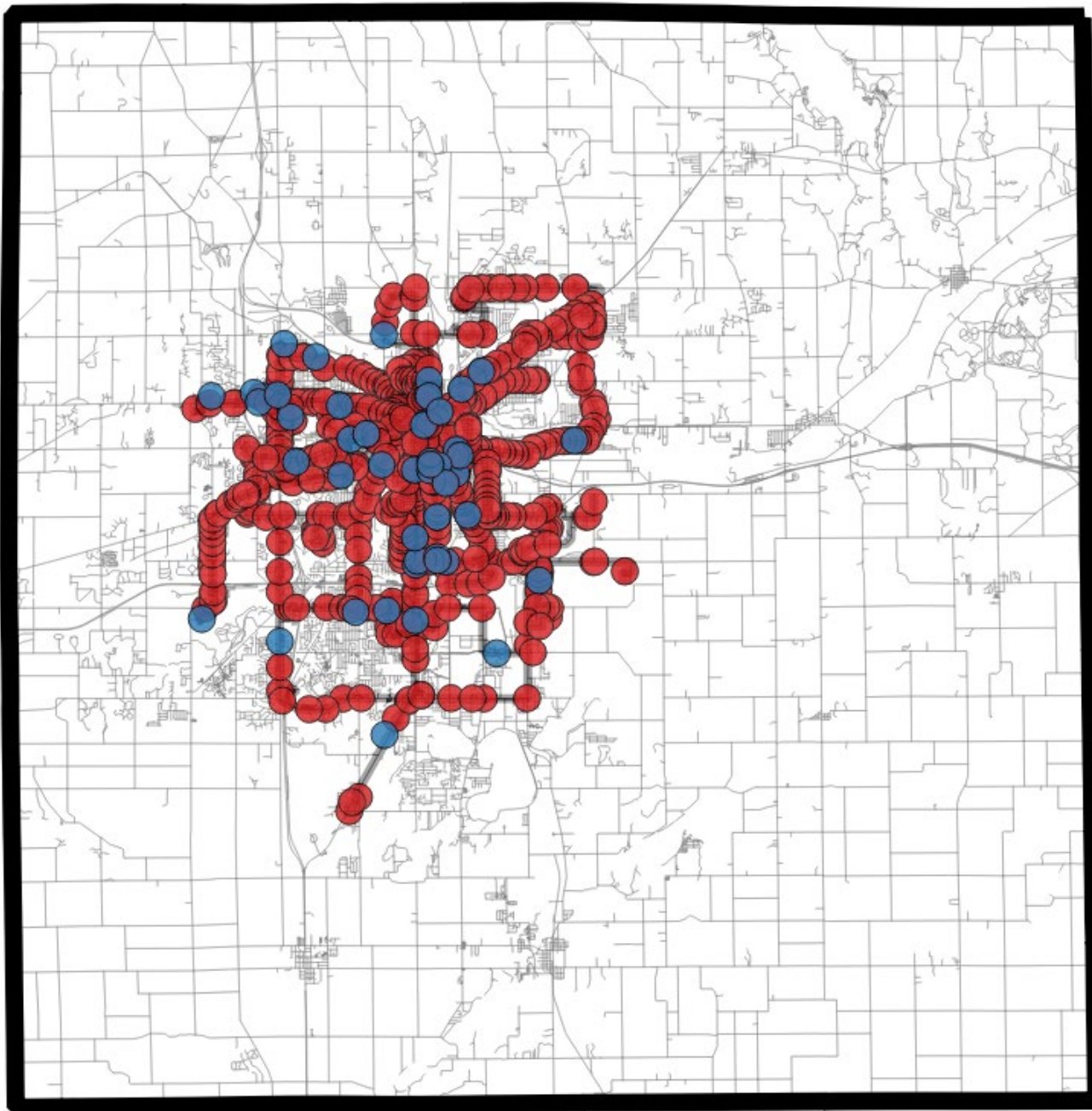
- Compliant (52)
- Not compliant (638)
- Kalamazoo County
- Kalamazoo Metro Bus Routes
- Roads



0 5 10 km

0 5 10 mi





## Kalamazoo Metro Bus Stops

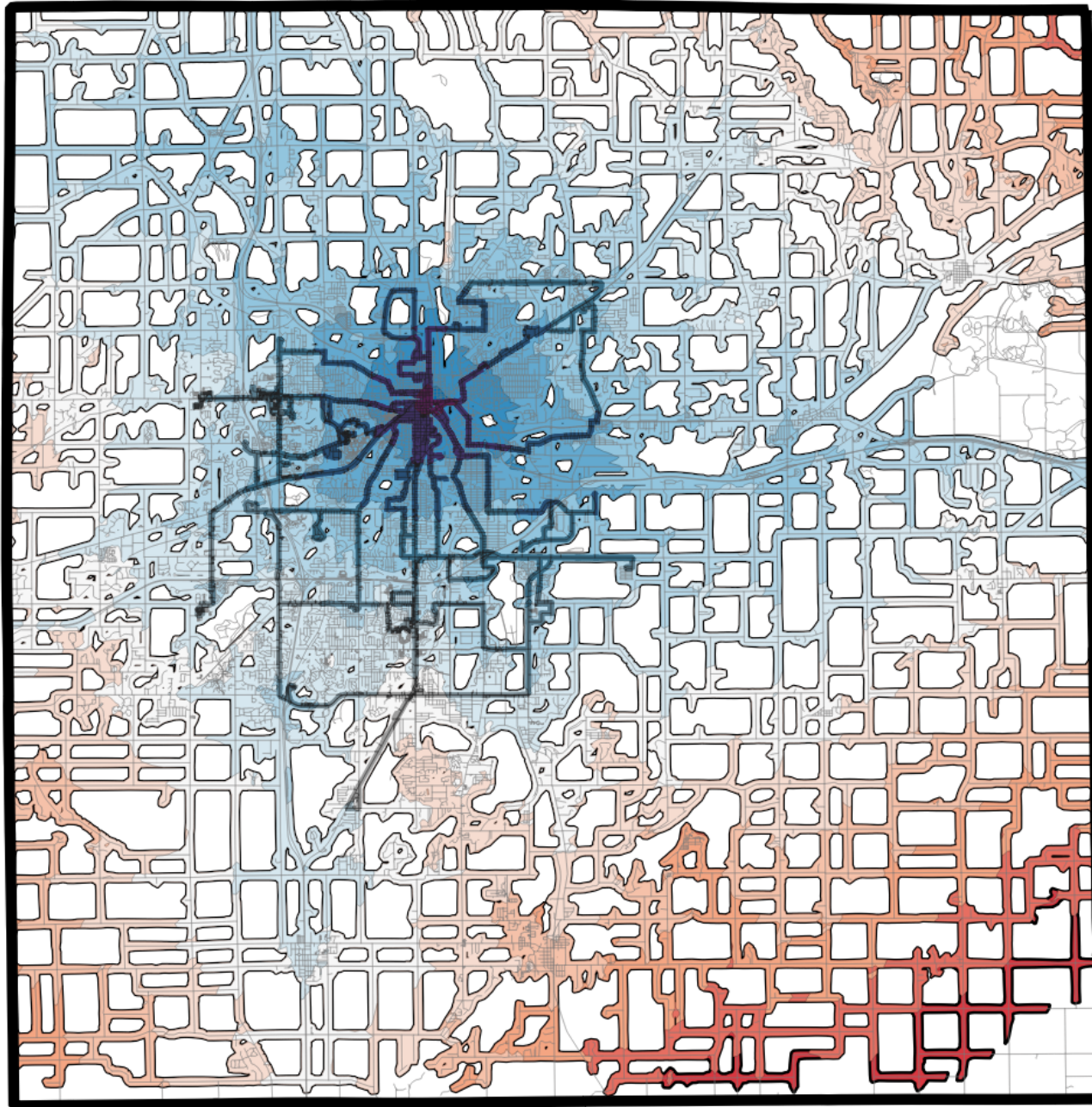
- Compliant (52)
- Not compliant (638)
- Kalamazoo County
- Kalamazoo Metro Bus Routes
- Roads



0 5 10 km

0 5 10 mi





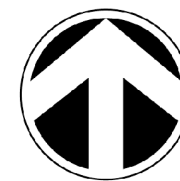
## Isochrones

- 5 minutes
- 10 minutes
- 15 minutes
- 20 minutes
- 25 minutes
- 30 minutes
- 35 minutes
- 50 minutes
- 45 minutes
- 50 minutes
- 55 minutes
- 60 minutes

— Kalamazoo Metro Bus Routes

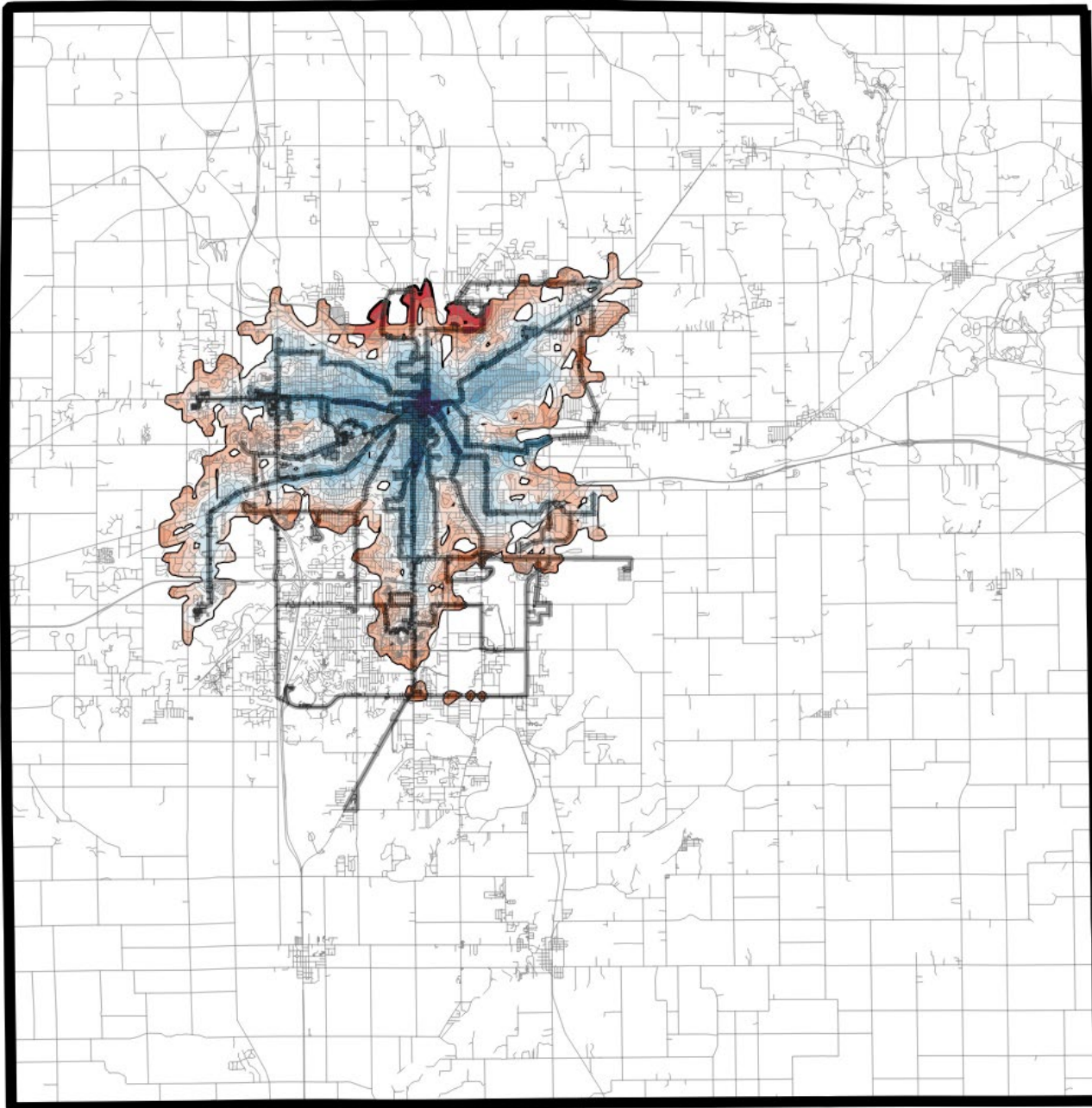
— Kalamazoo Roads

□ Kalamazoo County

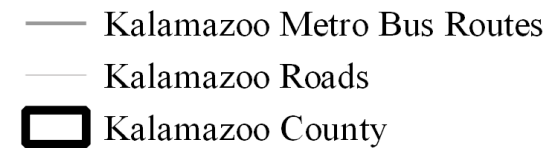
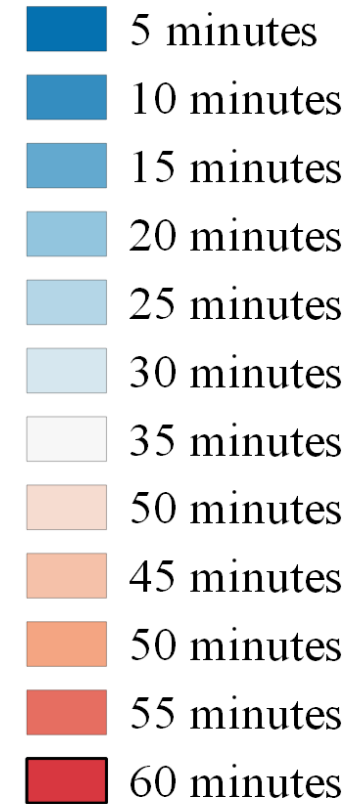


**Private automobile**



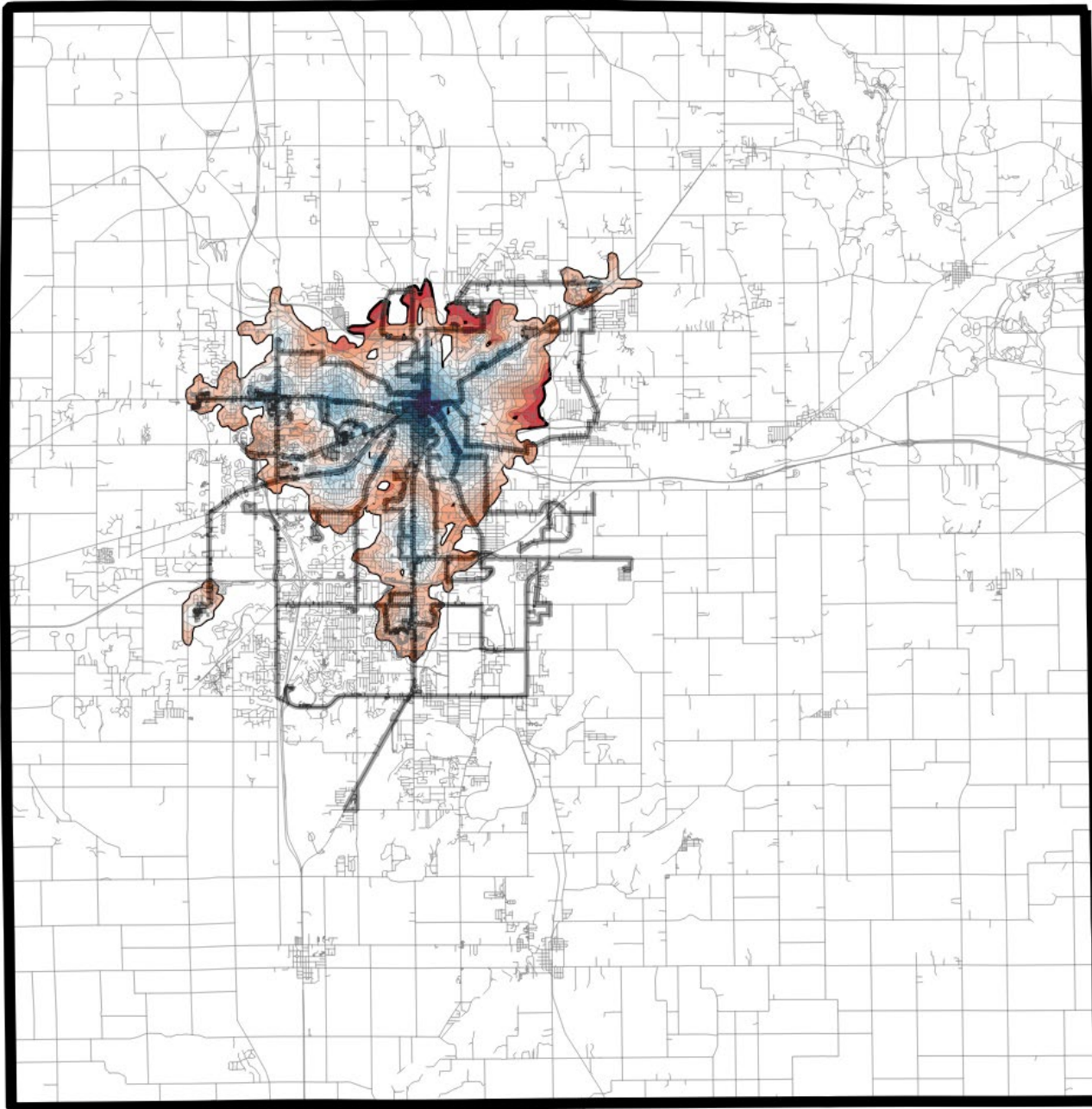


## Isochrones

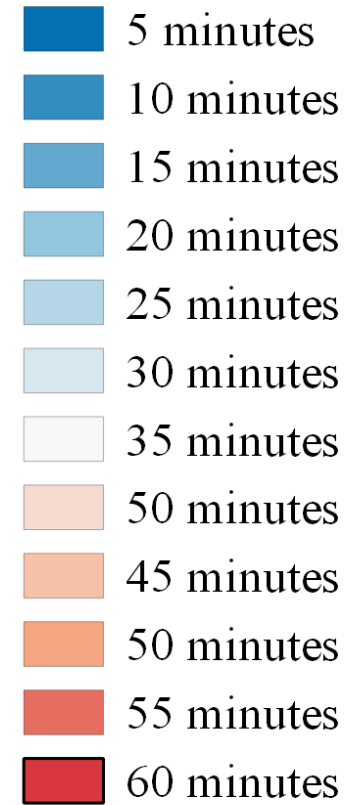


**Public transit +  
average pace  
walking**

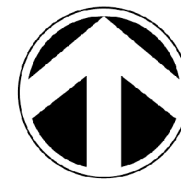




## Isochrones

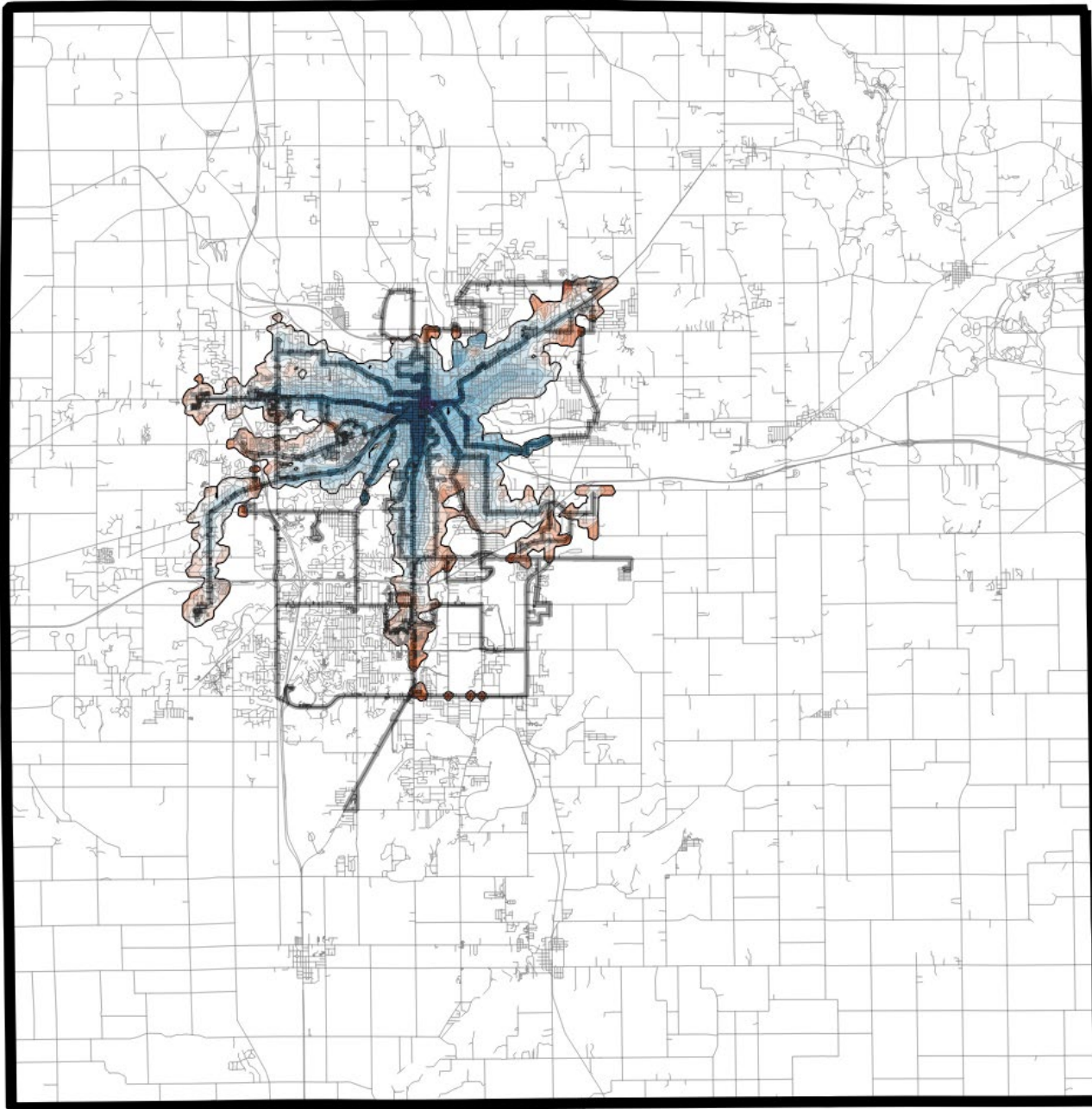


- Kalamazoo Metro Bus Routes
- Kalamazoo Roads
- Kalamazoo County

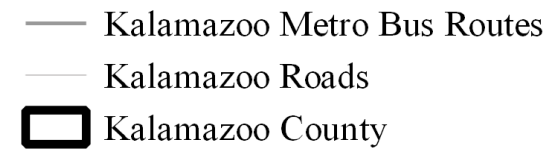
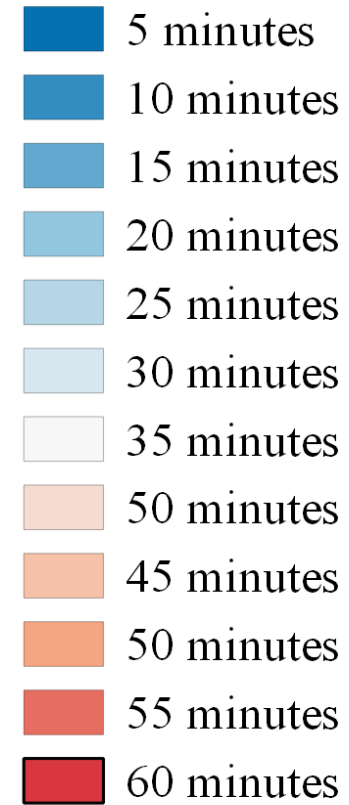


**Public transit +  
slower pace  
walking**

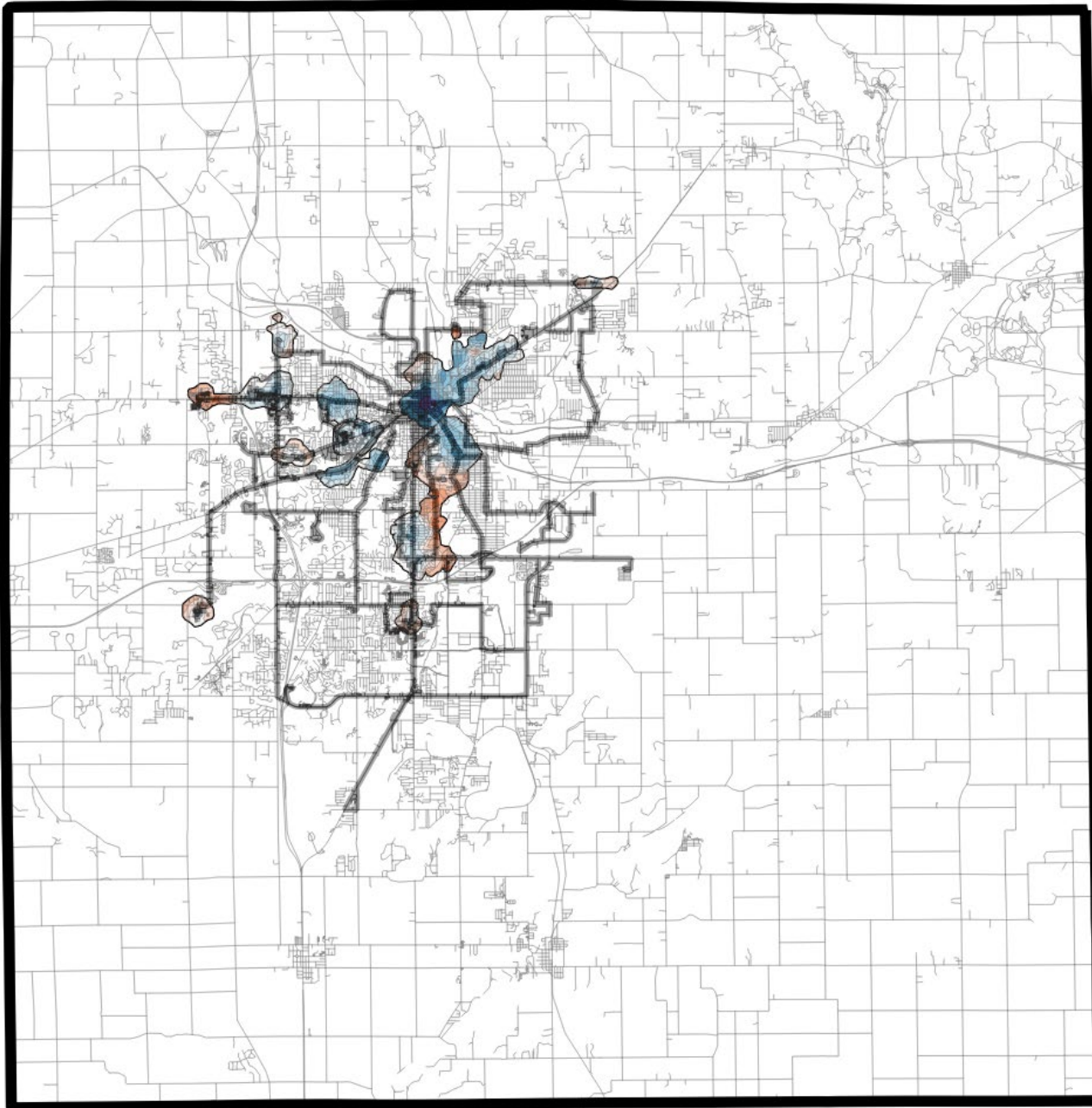




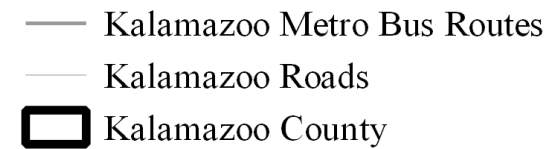
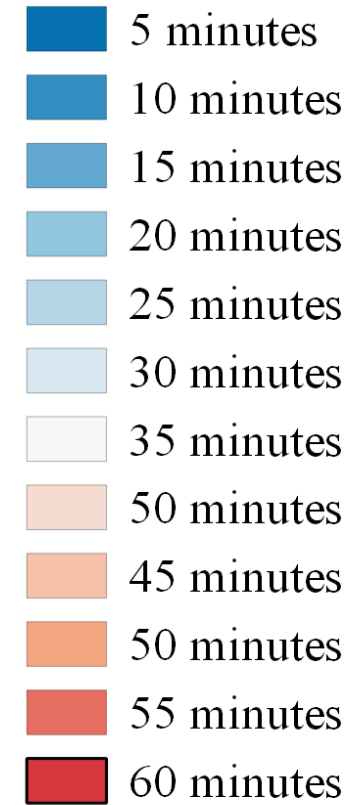
## Isochrones



**Accessible public  
transit + average  
pace walking**



## Isochrones



**Accessible public  
transit + slower  
pace walking**

